

1 Rings and Ideals

IDEALS. PRIME AND MAXIMAL IDEALS

Proposition 1.1. *There is a one-to-one order-preserving correspondence between the ideals $\mathfrak{b}$ of A which contain $\mathfrak{a}$, and the ideals $\bar{\mathfrak{b}}$ of $A/\mathfrak{a}$, given by $\mathfrak{b} = \phi^{-1}(\bar{\mathfrak{b}})$.*

Corollary 1.5. *Every non-unit of A is contained in a maximal ideal.*

Corollary 1.6. (i) *Let A be a ring and $\mathfrak{m} \neq (1)$ an ideal of A such that every $x \in A - \mathfrak{m}$ is a unit in A . Then A is a local ring and $\mathfrak{m}$ its maximal ideal.*

(ii) *Let A be a ring and $\mathfrak{m}$ a maximal ideal of A , such that every element of $1 + \mathfrak{m}$ is a unit in A , then A is a local ring.*

NILRADICAL AND JACOBSON RADICAL

We denote nilradical to be $\mathfrak{N}$ and Jacobson radical to be $\mathfrak{J}$.

Proposition 1.8. *The nilradical of A is the intersection of all the prime ideals of A .*

Sketch proof. Assume that $f \notin \mathfrak{N}$. Let Σ be a set of ideals $\mathfrak{a}$ with property $f^n \notin \mathfrak{a}$. Then Σ has maximal element which is prime. $\square$

Proposition 1.9. $x \in \mathfrak{J} \Leftrightarrow 1 - xy$ is a unit in A for all $y \in A$.

OPERATIONS ON IDEALS

Two ideals $\mathfrak{a}, \mathfrak{b}$ are said to be **coprime** if $\mathfrak{a} + \mathfrak{b} = 1$. Thus we have $\mathfrak{a} \cap \mathfrak{b} = \mathfrak{a}\mathfrak{b}$.

Proposition 1.10 (Chinese Remainder Theorem). *Define ϕ to be*

$$\phi : A \rightarrow \prod_{i=1}^n (A/\mathfrak{a}_i), \quad \phi(x) = (x + \mathfrak{a}_1, \dots, x + \mathfrak{a}_n).$$

- (i) *If $\mathfrak{a}_i$ and $\mathfrak{a}_j$ are coprime whenever $i \neq j$, then $\mathfrak{a}_1 \cdots \mathfrak{a}_n = \bigcap_{i=1}^n \mathfrak{a}_i$.*
- (ii) *ϕ is surjective $\Leftrightarrow \mathfrak{a}_i$ and $\mathfrak{a}_j$ are coprime whenever $i \neq j$.*
- (iii) *ϕ is injective $\Leftrightarrow \bigcap_{i=1}^n \mathfrak{a}_i = (0)$.*

Proposition 1.11. *Denote $\mathfrak{a}$ to be ideal and $\mathfrak{p}$ to be prime ideal.*

- (i) *If $\mathfrak{a} \subseteq \bigcup_{i=1}^n \mathfrak{p}_i$, then $\mathfrak{a} \subseteq \mathfrak{p}_i$ for some i .*
- (ii) *If $\mathfrak{p} \supseteq \bigcap_{i=1}^n \mathfrak{a}_i$, then $\mathfrak{p} \supseteq \mathfrak{a}_i$ for some i . If $\mathfrak{p} = \bigcap_{i=1}^n \mathfrak{a}_i$, then $\mathfrak{p} = \mathfrak{a}_i$ for some i .*

The **ideal quotient** of $\mathfrak{a}$ and $\mathfrak{b}$ is

$$(\mathfrak{a} : \mathfrak{b}) = \{x \in A : x\mathfrak{b} \subseteq \mathfrak{a}\}.$$

The annihilator of $\mathfrak{b}$ is $\text{Ann}(\mathfrak{b}) = (0 : \mathfrak{b})$.

The **radical** of $\mathfrak{a}$ is

$$\sqrt{\mathfrak{a}} = \{x \in A : x^n \in \mathfrak{a} \text{ for some } n > 0\}.$$

EXTENSION AND CONTRACTION

Let $f : A \rightarrow B$ be a ring homomorphism. Let $\mathfrak{a} \in A$ and $\mathfrak{b} \in B$. Define

$$\mathfrak{a}^e = Bf(\mathfrak{a}), \quad \mathfrak{b}^c = f^{-1}(\mathfrak{b})$$

to be the **extension** of $\mathfrak{a}$ and the **contraction** of $\mathfrak{b}$, respectively. If $\mathfrak{b}$ is prime, $\mathfrak{b}^c$ is prime; if $\mathfrak{a}$ is prime, $\mathfrak{a}^e$ need not be prime.

Proposition 1.17. $\mathfrak{a} \subseteq \mathfrak{a}^{ec}$, $\mathfrak{b} \supseteq \mathfrak{b}^{ce}$.

DEFINITIONS AND PROPOSITIONS FROM EXERCISES

Idempotent is an element e which satisfies $e^2 = e$.

A ring in which every element is an idempotent is a **Boolean ring**.

A topological space X is **reducible** if it can be written as a union $X = X_1 \cup X_2$ of two closed proper subsets X_1, X_2 of X . Equivalently, X is **irreducible** if every pair of non-empty open sets in X intersect, or if every non-empty open set is dense in X .

Let A be a ring and let $\text{Spec}(A)$ be the set of all prime ideals of A . For each subset E of A , let $V(E)$ denote the set of all prime ideals of A which contain E . By [Exercise 1.15](#), the sets $V(E)$ satisfy the axioms for closed sets in a topological space. The resulting topology on $\text{Spec}(A)$ is called **Zariski topology**, and $\text{Spec}(A)$ is called the **prime spectrum** of A .

For each $f \in A$, the complement of $V(f)$ in $\text{Spec}(A)$, denote to be X_f , is called **basic open set** of $\text{Spec}(A)$.

The subspace of $\text{Spec}(A)$ consisting of the maximal ideals of A with the induced topology, is called the **maximal spectrum** of A and is denoted by $\text{Max}(A)$.

Proposition ([Exercise 1.27](#)). *Let k be algebraically closed. Then the affine space k^n with its classical Zariski topology is homeomorphic to $\text{Max}(k[x_1, \dots, x_n])$.*

2 Modules

Proposition 2.1. (i) If $L \supseteq M \supseteq N$ are A -modules, then $(L/N)/(M/N) \cong L/M$.

(ii) If M_1, M_2 are submodules of M , then $(M_1 + M_2)/M_1 \cong M_2/(M_1 \cap M_2)$.

An A -module is **faithful** if $\text{Ann}(M) = 0$.

FINITELY GENERATED MODULES

Proposition 2.3. M is a finitely generated A -module $\Leftrightarrow M$ is isomorphic to a quotient of A^n for some integer $n > 0$. More precisely, n is the number of generators of M and may not be unique.

Proposition 2.4 (**Cayley–Hamilton theorem**). Let M be a finitely generated A -module, let $\mathfrak{a}$ be an ideal of A , and let ϕ be an A -module endomorphism of M such that $\phi(M) \subseteq \mathfrak{a}M$. Then ϕ satisfies an equation of the form

$$\phi^n + a_1\phi^{n-1} + \cdots + a_n = 0$$

where the a_i are in $\mathfrak{a}$.

Proposition 2.6 (**Nakayama's lemma**). Let M be a finitely generated A -module and $\mathfrak{a}$ an ideal of A contained in the Jacobson radical of A . Then $\mathfrak{a}M = M$ implies $M = 0$.

EXACT SEQUENCES

Proposition 2.10 (Snake lemma). Let

$$\begin{array}{ccccccc} 0 & \longrightarrow & M' & \xrightarrow{u} & M & \xrightarrow{v} & M'' \longrightarrow 0 \\ & & \downarrow f' & & \downarrow f & & \downarrow f'' \\ 0 & \longrightarrow & N' & \xrightarrow{u'} & N & \xrightarrow{v'} & N'' \longrightarrow 0 \end{array}$$

be a commutative diagram of A -modules and homomorphisms, with the rows exact. Then there exists an exact sequence

$$0 \longrightarrow \ker(f') \xrightarrow{\bar{u}} \ker(f) \xrightarrow{\bar{v}} \ker(f'') \xrightarrow{\delta} \text{coker}(f') \xrightarrow{\bar{u}'} \text{coker}(f) \xrightarrow{\bar{v}'} \text{coker}(f'') \longrightarrow 0.$$

Let C be a class of A -modules and let λ be a function on C with values in $\mathbb{Z}$. The function λ is **additive** if for each short exact sequence

$$0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$$

in which all terms belong to C , we have $\lambda(M') - \lambda(M) + \lambda(M'') = 0$.

TENSOR PRODUCT

Proposition 2.12 (Universal property). Let M, N be A -modules. The tensor product of M and N over A is an A -module $M \otimes_A N$ together with an A -bilinear map

$$f : M \times N \rightarrow M \otimes_A N, \quad (m, n) \mapsto m \otimes n,$$

such that, for every A -module P and every A -bilinear map $\tau : M \times N \rightarrow P$, there exists a unique A -linear map $\tilde{\tau} : M \otimes_A N \rightarrow P$ such that $\tau = \tilde{\tau} \circ f$. Equivalently, the following diagram commutes:

$$\begin{array}{ccc} M \times N & \xrightarrow{f} & M \otimes_A N \\ & \searrow \tau & \downarrow \tilde{\tau} \\ & & P \end{array}$$

Proposition 2.15. Let A, B be rings, let M be an A -module, P a B -module and N an (A, B) -bimodule. then

$$(M \otimes_A N) \otimes_B P \cong M \otimes_A (N \otimes_B P).$$

Let M be an A -module and $f : A \rightarrow B$ be a homomorphism of rings. Define $M_B = B \otimes_A M$.

Proposition 2.19. For an A -module N , the following are equivalent:

- (i) The functor $T_N : M \mapsto M \otimes_A N$ on the category of A -modules is exact.
- (ii) If $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ is any exact sequence of A -modules, then $0 \rightarrow M' \otimes N \rightarrow M \otimes N \rightarrow M'' \otimes N \rightarrow 0$ is exact.
- (iii) If $f : M' \rightarrow M$ is injective, then $f \otimes 1 : M' \otimes N \rightarrow M \otimes N$ is injective.
- (iv) If $f : M' \rightarrow M$ is injective and M, M' are finitely generated, then $f \otimes 1 : M' \otimes N \rightarrow M \otimes N$ is injective.

An A -module N satisfying these properties is said to be a *flat* A -module.

ALGEBRA

Let $f : A \rightarrow B$ be a ring homomorphism. If $a \in A$ and $b \in B$, define a product

$$ab = f(a)b.$$

Thus, B has an A -module structure. The ring B is said to be an *A -algebra*.

A ring homomorphism $f : A \rightarrow B$ is *finite*, if B is finitely generated A -module. We can also say B is a *finite* A -algebra, or B is *finite* over A .

The ring homomorphism f is *of finite type*, if B is a finitely generated A -algebra.

DEFINITIONS AND PROPOSITIONS FROM EXERCISES

A partially ordered set I is said to be a *directed* set if for each pair $i, j \in I$ there exists $k \in I$ such that $i \leq k$ and $j \leq k$.

Let A be a ring, let I be a directed set and let $(M_i)_{i \in I}$ be a family of A -modules indexed by I . For each pair $i, j \in I$ such that $i \leq j$, let $\mu_{ij} : M_i \rightarrow M_j$ be an A -homomorphism, and suppose:

- (1) μ_{ii} is the identity mapping of M_i , for all $i \in I$;
- (2) $\mu_{ik} = \mu_{jk} \circ \mu_{ij}$ whenever $i \leq j \leq k$.

Then the modules M_i and homomorphisms μ_{ij} are said to form a *direct system* $\mathbf{M} = (M_i, \mu_{ij})$ over the directed set I .

Let D be the submodule of $\bigoplus M_i$ generated by all elements of the form $x_i - \mu_{ij}(x_i)$ whenever $i \leq j$ and $x_i \in M_i$. The *direct limit* of the direct system $\mathbf{M}$ is defined to be

$$\varinjlim M_i = \bigoplus_{i \in I} M_i / D.$$

Let $\mu : \bigoplus M_i \rightarrow \varinjlim M_i$ be the projection of quotient module. Define μ_i to be the restriction of μ to M_i .

A ring A (not necessarily commutative) is **von Neumann regular** if for every $a \in A$, there is an $x \in A$ with $a = axa$. A commutative ring A is **absolutely flat** if every A -module is flat. By [Exercise 2.27](#), absolute flat and von Neumann regular are identical properties.

3 Rings and Modules of Fractions

MULTIPLICATIVELY CLOSED SETS

Proposition 3.3. *The operation S^{-1} is exact.*

Proposition 3.5. *Let M be an A -module. Then there exists a unique $S^{-1}A$ module isomorphism $f : S^{-1}A \otimes_A M \rightarrow S^{-1}M$.*

LOCAL PROPERTIES

Let P be a property of a ring A (or of an A -module M). Then the property P is said to be a **local property**, if A (or M) has $P \Leftrightarrow A_{\mathfrak{p}}$ (or $M_{\mathfrak{p}}$) has P for each prime ideal $\mathfrak{p}$ of A .

Proposition 3.8. *Let M, N be A -modules. Then the following are equivalent:*

- (i) $M = 0$;
- (ii) $M_{\mathfrak{p}} = 0$ for all prime ideals $\mathfrak{p}$ of A ;
- (iii) $M_{\mathfrak{m}} = 0$ for all maximal ideals $\mathfrak{m}$ of A ;

Proposition 3.10. *Let M be A -modules. Then the following are equivalent:*

- (i) M is a flat A -module;
- (ii) $M_{\mathfrak{p}}$ is a flat $A_{\mathfrak{p}}$ -module for each prime ideal $\mathfrak{p}$;
- (iii) $M_{\mathfrak{m}}$ is a flat $A_{\mathfrak{m}}$ -module for each maximal ideal $\mathfrak{m}$;

EXTENSIONS AND CONTRACTIONS OF FRACTIONS

If $\mathfrak{a}$ is an ideal in A , its extension $\mathfrak{a}^e$ in $S^{-1}A$ is $S^{-1}\mathfrak{a}$. Denote the contraction of $S^{-1}\mathfrak{a}$ in A to be $S(\mathfrak{a})$.

Proposition 3.11. (i) *Every ideal in $S^{-1}A$ is an extended ideal.*

- (ii) *If $\mathfrak{a}$ is an ideal in A , then $S(\mathfrak{a}) = \mathfrak{a}^{ec} = \bigcup_{s \in S} (\mathfrak{a} : s)$.*
- (iii) *$\mathfrak{a}$ is a contracted ideal $\Leftrightarrow$ no element of S is a zero-divisor in $A/\mathfrak{a}$.*
- (iv) *Them prime ideal of $S^{-1}A$ are in one-to-one correspondence ($\mathfrak{p} \leftrightarrow S^{-1}\mathfrak{p}$) with the prime ideal of A which don't meet S .*
- (v) *The operation S^{-1} commutes with formation of finite sums, products, intersections and radicals.*

Corollary 3.15. *If N, P are submodules of an A -module M and if P is finitely generated, then $S^{-1}(N : P) = (S^{-1}N : S^{-1}P)$.*

DEFINITIONS AND PROPOSITIONS FROM EXERCISES

A multiplicatively closed subset S of a ring A is said to be **saturated** if

$$xy \in S \Leftrightarrow x \in S \text{ and } y \in S.$$

Let M be an A -module. An element $x \in M$ is a **torsion** element of M if there exist a non-zero-divisor $a \in A$ such that $ax = 0$. All torsion elements of M form a submodule of M , which is called the **torsion submodule** of M and denote by $T(M)$. If $T(M) = 0$, the module M is said to be **torsion free**.

An A -module M is called **faithfully flat** if the tensor product functor $- \otimes_A M$ is exact and faithful. See [Exercise 3.16](#).

Let M be an A -module. The **support** of M is defined to be the set $\text{Supp}(M)$ of prime ideals $\mathfrak{p}$ such that $M_{\mathfrak{p}} \neq 0$.

$\text{Spec}(A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \otimes_A B)$ is called the **fiber** of $f^* : \text{Spec}(B) \rightarrow \text{Spec}(A)$ over $\mathfrak{p}$.

4 Primary Decomposition

PRIMARY IDEALS

An ideal $\mathfrak{q} \in A$ is **primary** if $\mathfrak{q} \neq A$, and if $x \notin \mathfrak{q} \Rightarrow y^n \in \mathfrak{q}$ for some $n > 0$. This also means $x \notin \sqrt{\mathfrak{q}} \Rightarrow y \in \mathfrak{q}$.

Proposition 4.4. *Let $\mathfrak{q}$ be a $\mathfrak{p}$ -primary ideal, $x \in A$ an element. Then*

- (i) *if $x \in \mathfrak{q}$ then $(\mathfrak{q} : x) = (1)$;*
- (ii) *if $x \notin \mathfrak{q}$ then $(\mathfrak{q} : x)$ is $\mathfrak{p}$ -primary;*
- (iii) *if $x \notin \mathfrak{p}$ then $(\mathfrak{q} : x) = \mathfrak{q}$;*

Proposition 4.8. *Let S be a multiplicatively closed subset of A , and let $\mathfrak{q}$ be a $\mathfrak{p}$ -primary ideal.*

- (i) *If $S \cap \mathfrak{p} \neq \emptyset$, then $S^{-1}\mathfrak{q} = S^{-1}A$.*
- (ii) *If $S \cap \mathfrak{p} = \emptyset$, then $S^{-1}\mathfrak{q}$ is $S^{-1}\mathfrak{p}$ -primary and its contraction in A is $\mathfrak{q}$.*

A **primary decomposition** of an ideal $\mathfrak{a}$ in A is an expression $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$.

If $\sqrt{\mathfrak{q}_i}$ are all distinct and $\mathfrak{q}_i \not\supseteq \bigcap_{j \neq i} \mathfrak{q}_j$, the primary decomposition is said to be **irredundant**.

UNIQUENESS

Proposition 4.5 (1st uniqueness theorem). *Let $\mathfrak{a}$ be a decomposable ideal and let $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ be a minimal primary decomposition of $\mathfrak{a}$. Let $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$. Then $\mathfrak{p}_i$ are precisely the prime ideals which occur in the set of ideals $\sqrt{(\mathfrak{a} : x)}$, and hence are independent of the particular decomposition of $\mathfrak{a}$.*

The prime ideals $\mathfrak{p}_i$ are said to belong to $\mathfrak{a}$, or to be **associated** with $\mathfrak{a}$. The minimal elements of the set $\{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$ are called the minimal or **isolated** prime ideals belonging to $\mathfrak{a}$. The others are called **embedded** prime ideals.

Proposition 4.9. *Let S be a multiplicatively closed subset of A and let $\mathfrak{a}$ be a decomposable ideal. Let $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ be a minimal primary decomposition of $\mathfrak{a}$. Let $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$ and suppose the $\mathfrak{q}_i$ numbered so that S meets $\mathfrak{p}_{m+1}, \dots, \mathfrak{p}_n$ but not $\mathfrak{p}_1, \dots, \mathfrak{p}_m$. Then we have minimal primary decompositions*

$$S^{-1}\mathfrak{a} = \bigcap_{i=1}^m S^{-1}\mathfrak{q}_i, \quad S(\mathfrak{a}) = \bigcap_{i=1}^m \mathfrak{q}_i.$$

A set Σ of prime ideals belonging to $\mathfrak{a}$ is said to be **isolated** if it satisfies the following condition: if $\mathfrak{p}'$ is a prime ideal belonging to $\mathfrak{a}$ and $\mathfrak{p}' \subseteq \mathfrak{p}$ for some $\mathfrak{p} \in \Sigma$, then $\mathfrak{p}' \in \Sigma$. Thus, every isolated set contains an isolated prime ideal.

Proposition 4.10 (2nd uniqueness theorem). *Let $\mathfrak{a}$ be a decomposable ideal, let $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ be a minimal primary decomposition of $\mathfrak{a}$, and let $\{\mathfrak{p}_{i_1}, \dots, \mathfrak{p}_{i_m}\}$ be an isolated set of prime ideals of $\mathfrak{a}$. Then the ideal $\mathfrak{q}_{i_1} \cap \dots \cap \mathfrak{q}_{i_m}$ is independent of the decomposition.*

Corollary 4.11. *The isolated primary components are uniquely determined by $\mathfrak{a}$.*

DEFINITIONS AND PROPOSITIONS FROM EXERCISES

Let A be a ring and $\mathfrak{p}$ a prime ideal of A . The n th **symbolic power** of $\mathfrak{p}$ is defined to be the ideal $\mathfrak{p}^{(n)} = S_{\mathfrak{p}}(\mathfrak{p}^n)$.

5 Integral Dependence and Valuations

INTEGRAL DEPENDENCE

Proposition 5.1. *The following are equivalent:*

- (i) $x \in B$ is integral over A ;
- (ii) $A[x]$ is a finitely generated A -module;
- (iii) $A[x]$ is contained in a subring C of B such that C is a finitely generated A -module;
- (iv) There exist a faithful $A[x]$ -module M which is finitely generated as an A -module.

The set $\bar{A}$ of elements of B which are integral over A is called the **integral closure** of A in B . If $\bar{A} = A$, then A is said to be **integrally closed in B** . If $\bar{A} = B$, the ring B is said to be **integral over A** .

THE GOING-UP THEOREM

Proposition 5.7. *Let $A \subseteq B$ be integral domains, B integral over A . Then B is a field iff A is a field.*

Corollary 5.9. *Let $A \subseteq B$ be rings, B integral over A . Let $\mathfrak{q}_1, \mathfrak{q}_2$ be prime ideals of B such that $\mathfrak{q}_1 \subseteq \mathfrak{q}_2$ and $\mathfrak{q}_1^e = \mathfrak{q}_2^e = \mathfrak{p}$. Then $\mathfrak{q}_1 = \mathfrak{q}_2$.*

Theorem 5.10 (Lying over property). *Let $A \subseteq B$ be rings, B integral over A , and let $\mathfrak{p}$ be a prime ideal of A . Then there exists a prime ideal $\mathfrak{q}$ of B such that $\mathfrak{q} \cap A = \mathfrak{p}$.*

Theorem 5.11 (Going-up theorem). *Let $A \subseteq B$ be rings, B integral over A . Let $\mathfrak{p}_1 \subseteq \cdots \subseteq \mathfrak{p}_n$ be a chain of prime ideals of A and $\mathfrak{q}_1 \subseteq \cdots \subseteq \mathfrak{q}_m$ ($m < n$) a chain of prime ideals of B such that $\mathfrak{q}_i \cap A = \mathfrak{p}_i$ for $1 \leq i \leq m$. Then the chain $\mathfrak{q}_1 \subseteq \cdots \subseteq \mathfrak{q}_m$ can be extended to a chain $\mathfrak{q}_1 \subseteq \cdots \subseteq \mathfrak{q}_n$ such that $\mathfrak{q}_i \cap A = \mathfrak{p}_i$ for $1 \leq i \leq n$.*

THE GOING-DOWN THEOREM

Proposition 5.12. *Let $A \subseteq B$ be rings, C the integral closure of A in B . Let S be a multiplicatively closed subset of A . Then $S^{-1}C$ is the integral closure of $S^{-1}A$ in $S^{-1}B$.*

An integral domain is said to be **integrally closed** (without qualification) if it is integrally closed in its field of fractions.

Proposition 5.13. *Let A be an integral domain. Then the following are equivalent:*

- (i) A is integrally closed;
- (ii) $A_{\mathfrak{p}}$ is integrally closed, for each prime ideal $\mathfrak{p}$;
- (iii) $A_{\mathfrak{m}}$ is integrally closed, for each maximal ideal $\mathfrak{m}$.

Theorem 5.16 (Going-down theorem). *Let $A \subseteq B$ be integral domains, A integrally closed, B integral over A . Let $\mathfrak{p}_1 \supseteq \cdots \supseteq \mathfrak{p}_n$ be a chain of prime ideals of A and $\mathfrak{q}_1 \supseteq \cdots \supseteq \mathfrak{q}_m$ ($m < n$) a chain of prime ideals of B such that $\mathfrak{q}_i \cap A = \mathfrak{p}_i$ for $1 \leq i \leq m$. Then the chain $\mathfrak{q}_1 \supseteq \cdots \supseteq \mathfrak{q}_m$ can be extended to a chain $\mathfrak{q}_1 \supseteq \cdots \supseteq \mathfrak{q}_n$ such that $\mathfrak{q}_i \cap A = \mathfrak{p}_i$ for $1 \leq i \leq n$.*

VALUATION RINGS

Let B be an integral domain. B is a **valuation ring** of $\text{Frac}(B)$ if, for each $x \neq 0$, either $x \in B$ or $x^{-1} \in B$ (or both).

Proposition 5.18. *Let B be a valuation ring, $K = \text{Frac}(B)$ its field of fractions. Then*

- (i) B is a local ring;
- (ii) If B' is a ring such that $B \subseteq B' \subseteq K$, then B' is a valuation ring of K ;
- (iii) B is integrally closed.

Theorem 5.21. *Let K be a field, Ω an algebraically closed field. Let Σ be the set of all pairs (A, f) , where A is a subring of K and $f : A \rightarrow \Omega$ a homomorphism. Define a partial order of Σ to be:*

$$(A, f) \leq (B, g) \Leftrightarrow A \subseteq B \text{ and } g|_A = f.$$

Then the maximal element of Σ is a valuation ring of K .

DEFINITIONS AND PROPOSITIONS FROM EXERCISES

A ring homomorphism $f : A \rightarrow B$ is said to have the **going-up property** if for $\mathfrak{p}_1 \subseteq \mathfrak{p}_2$ and $\mathfrak{p}_1 = f^{-1}(\mathfrak{q}_1)$, there exist a $\mathfrak{q}_2$ such that $\mathfrak{q}_1 \subseteq \mathfrak{q}_2$ and $f^{-1}(\mathfrak{q}_2) = \mathfrak{p}_2$. Reversing the inclusions of prime ideals we have the **going-down property**.

Proposition (Noether normalization lemma). *Let k be a field and let $A \neq 0$ be a finitely generated k -algebra. Then there exist elements $y_1, \dots, y_r \in A$ which are algebraically independent over k and such that A is integral over $k[y_1, \dots, y_r]$.*

A ring A satisfied the following equivalent property is a **Jacobson ring**.

- (i) Every prime ideal in A is an intersection of maximal ideals.
- (ii) In every homomorphic image of A the nilradical is equal to the Jacobson radical.
- (iii) Every prime ideal in A which is not maximal is equal to the intersection of the prime ideals which contain it strictly.

Let A be a valuation ring of a field K . The group U of units of A is a subgroup of the multiplicative group K^* of K . Let $\Gamma = K^*/U$. If $\xi, \iota \in \Gamma$ are represented by $x, y \in K$, define $\xi \geq \iota$ to mean $xy^{-1} \in A$. Then Γ is a totally ordered abelian group. It is called the **value group** of A .

Conversely, let Γ be a totally ordered abelian group. A **valuation** of K with values in Γ is a mapping $v : K^* \rightarrow \Gamma$ such that

- (1) $v(xy) = v(x) + v(y)$,
- (2) $v(x + y) \geq \min(v(x), v(y))$,

for all $x, y \in K^*$. Then the set of elements $x \in K^*$ such that $v(x) \geq 0$ is a valuation ring of K . This ring is called the **valuation ring** of v , and the subgroup $v(K^*)$ of Γ is the **value group** of v . Thus the concepts of valuation ring and valuation are equivalent.

6 Chain Conditions

Let Σ be a partially ordered set. Σ said to be satisfying the **ascending chain condition** (a.c.c) if, for every increasing sequence $x_1 \leq x_2 \leq \dots$ in Σ there exists n such that $x_n = x_{n+1} = \dots$.

Similarly, Σ said to be satisfying the **descending chain condition** (d.c.c) if, for every decreasing sequence $x_1 \geq x_2 \geq \dots$ in Σ there exists n such that $x_n = x_{n+1} = \dots$.

A module satisfying the ascending chain condition is said to be **Noetherian**. A module satisfying the descending chain condition is said to be **Artinian**.

Proposition 6.2. *M is a Noetherian A -module $\Leftrightarrow$ every submodule of M is finitely generated.*

Proposition 6.3. *Let $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ be an exact sequence of A -modules. Then*

(i) *M is Noetherian $\Leftrightarrow M'$ and M'' are Noetherian.*

(ii) *M is Artinian $\Leftrightarrow M'$ and M'' are Artinian.*

Proposition 6.5. *Let A be a Noetherian (resp. Artinian) ring, M a finitely generated A -module. Then M is Noetherian (resp. Artinian).*

A **chain** of submodules of a module M is a sequence (M_i) of submodules of M such that

$$M = M_0 \supsetneq M_1 \supsetneq \dots \supsetneq M_n = 0.$$

The **length** of the chain is n . A **composition series** of M is a maximal chain, that is one which no extra submodules can be inserted, or equivalently M_{i-1}/M_i is simple.

Proposition 6.7. *Suppose that M has a composition series of length n . Then every composition series of M has length n , and every chain can be extended to a composition series.*

The length of a composition series of M is called the **length** of M , denoted as $l(M)$.

Proposition 6.10. *For a vector space,*

$$\text{finite dimension} \iff \text{finite length} \iff \text{a.c.c.} \iff \text{d.c.c..}$$

If the conditions are satisfied, length equals dimension.

DEFINITIONS AND PROPOSITIONS FROM EXERCISES

A topological space X is said to be **Noetherian** if the open subsets of X satisfy the ascending chain condition. Or equivalently, the closed subsets satisfy descending chain condition.

7 Noetherian Rings

Proposition 7.1. *Any homomorphic image of Noetherian ring is Noetherian.*

Proposition 7.2. *Let A be a subring of B ; suppose that A is Noetherian and that B is finitely generated as an A -module. Then B is Noetherian.*

Proposition 7.3. *If A is Noetherian and S is any multiplicatively closed subset of A , then $S^{-1}A$ is Noetherian.*

Theorem 7.5 (Hilbert's basis theorem). *If A is Noetherian, then the polynomial ring $A[x]$ is Noetherian.*

Proposition 7.8. *Let $A \subseteq B \subseteq C$ be rings. Suppose that A is Noetherian, that C is finitely generated as an A -algebra and that C is either (i) finitely generated as a B -module or (ii) integral over B . Then B is finitely generated as an A -algebra.*

Proposition 7.9 (Zariski's lemma). *Let k be a field, E a finitely generated k -algebra. If E is a field then it is a finite algebraic extension of k .*

PRIMARY DECOMPOSITION IN NOETHERIAN RINGS

Proposition 7.11. *In a Noetherian ring every ideal is a finite intersection of irreducible ideals.*

Proposition 7.12. *In a Noetherian ring every irreducible ideal is primary.*

Proposition 7.14. *In a Noetherian ring every ideal contains a power of its radical.*

Proposition 7.15. *In a Noetherian ring the nilradical is nilpotent.*

Proposition 7.16. *Let A be a Noetherian ring, $\mathfrak{m}$ a maximal ideal of A , $\mathfrak{q}$ any ideal of A . Then the following are equivalent:*

- (i) $\mathfrak{q}$ is $\mathfrak{m}$ -primary;
- (ii) $\sqrt{\mathfrak{q}} = \mathfrak{m}$;
- (iii) $\mathfrak{m}^n \subseteq \mathfrak{q} \subseteq \mathfrak{m}$ for some $n > 0$.

Proposition 7.17. *Let $\mathfrak{a} \neq (1)$ be an ideal in a Noetherian ring. Then the prime ideals which belong to $\mathfrak{a}$ are precisely the prime ideals which occur in the set of ideals $(\mathfrak{a} : x)$.*

8 Artinian Rings

Proposition 8.1. *In an Artinian ring every prime ideal is maximal.*

Proposition 8.3. *An Artinian ring has only a finite number of maximal ideals.*

Proposition 8.4. *In an Artinian ring the nilradical is nilpotent.*

The **dimension** of A is the supremum of the length of all chains of prime ideals in A .

Proposition 8.5. *A ring A is Artinian $\Leftrightarrow A$ is Noetherian and $\dim(A) = 0$.*

Proposition 8.6. *Let A be a Noetherian local ring, $\mathfrak{m}$ its maximal ideal. Then exactly one of the following statements is true:*

- (i) $\mathfrak{m}^n \neq \mathfrak{m}^{n+1}$ for all n ;
- (ii) $\mathfrak{m}^n = 0$ for some n . In this case A is an Artinian local ring.

Proposition 8.7 (Structure theorem of Artinian rings). *An Artinaian ring A is uniquely (up to isomorphism) a finite direct product of Artinian local rings.*

Proposition 8.8. *Let A be an Artinian local ring. Then the following are equivalent:*

- (i) *every ideal in A is principal;*
- (ii) *the maximal ideal $\mathfrak{m}$ is principal;*
- (iii) $\dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2) \leq 1$.

9 Discrete Valuation Rings and Dedekind Domain

Proposition 9.1. *Let A be a Noetherian domain of dimension 1. Then every non-zero ideal $\mathfrak{a}$ in A can be uniquely expressed as a product of primary ideals whose radicals are all distinct.*

DISCRETE VALUATION RINGS

Let K be a field. A **discrete valuation** on K is a mapping v of K^* onto $\mathbb{Z}$ such that

- (1) $v(xy) = v(x) + v(y)$;
- (2) $v(x + y) \geq \min(v(x), v(y))$.

The set consisting of all $x \in K^*$ such that $v(x) \geq 0$ called the **valuation ring** of v .

An integral domain A is a **discrete valuation ring** if there is a discrete valuation v of its field of fractions K such that A is the valuation ring of v .

Proposition 9.2. *Let A be a Noetherian local domain of dimension 1, $\mathfrak{m}$ its maximal ideal. Then the following are equivalent:*

- (i) A is a discrete valuation ring;
- (ii) A is integrally closed;
- (iii) $\mathfrak{m}$ is a principal ideal;
- (iv) $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = 1$;
- (v) Every non-zero ideal is a power of $\mathfrak{m}$;
- (vi) There exists $x \in A$ such that every non-zero ideal is of the form (x^k) for $k \geq 0$.

DEDEKIND DOMAINS

Theorem 9.3. *Let A be a Noetherian local domain of dimension 1. Then the following are equivalent:*

- (i) A is integrally closed;
- (ii) Every primary ideal in A is a prime power;
- (iii) Every local ring $A_{\mathfrak{p}}$ with $\mathfrak{p} \neq 0$ is a discrete valuation ring.

A ring satisfying the conditions is called a **Dedekind domain**.

FRACTIONAL IDEALS

Let A be an integral domain, K its field of fractions. An A -submodule M of K is a **fractional ideal** of A if $xM \subseteq A$ for some $x \neq 0$.

An A -submodule M of K is an **invertible ideal** if there exists a submodule N of K such that $MN = A$. Then $N = (A : M)$, and M is finitely generated fractional ideal.

Proposition 9.6. *For a fractional ideal M , the following are equivalent:*

- (i) M is invertible;

(ii) M is finitely generated and , for each prime ideal $\mathfrak{p}$, $M_{\mathfrak{p}}$ is invertible;

(iii) M is finitely generated and , for each maximal ideal $\mathfrak{m}$, $M_{\mathfrak{m}}$ is invertible;

Theorem 9.8. *Let A be an integral domain. Then A is a Dedekind domain $\Leftrightarrow$ every non-zero fractional ideal of A is invertible.*

10 Completion

TOPOLOGIES AND COMPLETIONS

A **topological abelian group** G is both a topological space and an abelian group, with two structural mappings

$$\begin{aligned} G \times G &\rightarrow G : (x, y) \mapsto x + y \\ G &\rightarrow G : x \mapsto -x \end{aligned}$$

being continuous.

If $\{0\}$ is closed in G , then G is Hausdorff.

Since $T_a(x) = x + a$ is a homeomorphism, the topology of G is uniquely determined by the neighborhoods of 0 in G .

Proposition 10.1. *Let H be the intersection of all neighborhoods of 0 in G . Then*

- (i) H is a subgroup.
- (ii) H is the closure of $\{0\}$.
- (iii) G/H is Hausdorff.
- (iv) G is Hausdorff $\Leftrightarrow H = 0$.

Assume that $0 \in G$ has a countable fundamental system of neighborhoods. A **Cauchy sequence** (x_ν) of G is that, for any neighborhood U of 0, there exists an integer $s(U)$ such that $x_m - x_n \in U$ for all $m, n \geq s(U)$.

Let $C(G)$ be the set of all Cauchy sequences in G . Define an equivalence relation $\sim$ on $C(G)$ where $(x_\nu) \sim (y_\nu)$ means $x_n - y_n \rightarrow 0$ in G . The **completion** $\hat{G}$ of G is the set of all equivalence classes in $C(G)/\sim$.

For each $x \in G$, define a homomorphism $\phi : G \rightarrow \hat{G}$ maps to a constant sequence (x) . Then ϕ is injective iff G is Hausdorff.

Assume further that $0 \in G$ has a fundamental system of neighborhoods consisting of a sequence of subgroups

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_n \supseteq \cdots$$

For any $(x_\nu) \in C(G)$ and a fixed n , the image sequence $(\xi_\nu + G_n)$ in G/G_n is ultimately constant. Let ξ_n be this ultimate constant value.

As n varies, the projection $\theta_{n+1} : G/G_{n+1} \rightarrow G/G_n$ satisfies $\theta_{n+1}(\xi_{n+1}) = \xi_n$ for all n . The groups G/G_n combined with maps θ_n form an **inverse system**, and the group of all such coherent sequences (ξ_ν) is called the **inverse limit**, denoted as

$$\hat{G} = \varprojlim G/G_n.$$

A direct (or inverse) system is called **injective/surjective system**, if every transition map μ_{ij} is injective/surjective.

Proposition 10.2. *If $0 \rightarrow \{A_n\} \rightarrow \{B_n\} \rightarrow \{C_n\} \rightarrow 0$ is an exact sequence of inverse system, then*

$$0 \rightarrow \varprojlim A_n \rightarrow \varprojlim B_n \rightarrow \varprojlim C_n$$

is exact. If $\{A_n\}$ is surjective, then

$$0 \rightarrow \varprojlim A_n \rightarrow \varprojlim B_n \rightarrow \varprojlim C_n \rightarrow 0$$

is exact.

Let $A = \prod_{i=1}^{\infty} A_n$ and define $d^A : A \rightarrow A$ by $d^A(a_n) = a_n - \theta_{n+1}(a_{n+1})$. Then $\ker(d^A) \cong \varprojlim A_n$. Similar to B and C , we have a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \\ & & \downarrow d^A & & \downarrow d^B & & \downarrow d^C \\ 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \end{array}$$

By [Snake lemma](#), we have an exact sequence

$$0 \longrightarrow \ker(d^A) \longrightarrow \ker(d^B) \longrightarrow \ker(d^C) \longrightarrow \operatorname{coker}(d^A) \longrightarrow \operatorname{coker}(d^B) \longrightarrow \operatorname{coker}(d^C) \longrightarrow 0.$$

and the $\operatorname{coker}(d^A)$ is usually denoted by $\varprojlim^1 A_n$.

Corollary 10.3. *Let $0 \rightarrow G' \rightarrow G \rightarrow G'' \rightarrow 0$ be an exact sequence of groups. Let G have the topology defined by a sequence $\{G_n\}$ of subgroups, and give G' and G'' the topologies by the contraction and extension of $\{G_n\}$, respectively. Then*

$$0 \rightarrow \hat{G}' \rightarrow \hat{G} \rightarrow \hat{G}'' \rightarrow 0$$

is exact.

Corollary 10.4. $\hat{G}_n$ is a subgroup of $\hat{G}$ and $\hat{G}/\hat{G}_n \cong G/G_n$.

Corollary 10.5. $\hat{\hat{G}}_n \cong \hat{G}_n$.

If $G \rightarrow \hat{G}$ is an isomorphism, G is said to be [complete](#).

Let A be a ring, $\mathfrak{a}$ its ideal. The topology defined by $G_n = \mathfrak{a}^n$ ($G_0 = A$) is called the [\$\mathfrak{a}\$ -adic topology](#), and the ring A combined with this topology is a [topological ring](#).

FILTRATIONS

A chain $M = M_0 \supseteq M_1 \supseteq \cdots \supseteq M_n \cdots$ is called a [filtration](#) of M , and denoted by (M_n) . It is an [\$\mathfrak{a}\$ -filtration](#) if $\mathfrak{a}M_n \subseteq M_{n+1}$ for all n , and a [stable \$\mathfrak{a}\$ -filtration](#) if $\mathfrak{a}M_n = M_{n+1}$ for all sufficiently large n .

The [\$\mathfrak{a}\$ -adic completion](#) of M is defined to be

$$\hat{M} = \varprojlim M/\mathfrak{a}^n M.$$

GRADED RINGS AND MODULES

A [graded ring](#) is a ring A together with a family $(A_n)_{n \geq 0}$ of additive subgroups of A , such that $A = \bigoplus_{n=0}^{\infty} A_n$ and $A_m A_n \subseteq A_{m+n}$ for all $m, n \geq 0$.

Proposition 10.7. *A graded ring A is Noetherian, iff A_0 is Noetherian and A is finitely generated as an A_0 -algebra.*

Proposition 10.9 (Artin-Rees lemma). *Let A be a Noetherian ring, $\mathfrak{a}$ an ideal in A , M a finitely-generated A -module, (M_n) a stable $\mathfrak{a}$ -filtration of M . If M' is a submodule of M , then $(M' \cap M_n)$ is a stable $\mathfrak{a}$ -filtration of M' .*

Proposition 10.12. *Let $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ be an exact sequence of finitely-generated modules over a Noetherian ring A , then the sequence of $\mathfrak{a}$ -adic completion $0 \rightarrow \hat{M}' \rightarrow \hat{M} \rightarrow \hat{M}'' \rightarrow 0$ is exact.*

Proposition 10.13. *Let $f : \hat{A} \otimes_A M \rightarrow \hat{A} \otimes_A \hat{M} \rightarrow \hat{A} \otimes_{\hat{A}} \hat{M} \rightarrow \hat{M}$. If M is finitely-generated A -module, f is surjective. If moreover A is Noetherian, f is an isomorphism.*

Proposition 10.14. *If A is a Noetherian ring, the $\mathfrak{a}$ -adic completion $\hat{A}$ is a flat A -algebra.*

Proposition 10.15. *If A is a Noetherian ring, $\hat{A}$ its $\mathfrak{a}$ -adic completion, then*

- (i) $\hat{\mathfrak{a}} = \hat{A}\mathfrak{a} \cong \hat{A} \otimes_A \mathfrak{a}$;
- (ii) $\widehat{(\mathfrak{a}^n)} = (\hat{\mathfrak{a}})^n$;
- (iii) $\hat{\mathfrak{a}}^n / \hat{\mathfrak{a}}^{n+1} \cong \mathfrak{a}^n / \mathfrak{a}^{n+1}$;
- (iv) $\hat{\mathfrak{a}}$ is contained in the Jacobson radical of $\hat{A}$.

Proposition 10.16. *Let A be a Noetherian local ring, $\mathfrak{m}$ its maximal ideal. Then the $\mathfrak{m}$ -adic completion $\hat{A}$ of A is a local ring with maximal ideal $\hat{\mathfrak{m}}$.*

Proposition 10.17 (Krull intersection theorem). *Let A be a Noetherian ring, $\mathfrak{a}$ an ideal, M a finitely-generated A -module and $\hat{M}$ the $\mathfrak{a}$ -adic completion of M . Then the kernel $\bigcap_{n=1}^{\infty} \mathfrak{a}^n M$ of $M \rightarrow \hat{M}$ consists of all $x \in M$ annihilated by some element of $1 + \mathfrak{a}$.*

THE ASSOCIATED GRADED RING

Let A be a ring and $\mathfrak{a}$ an ideal of A . The **associated graded ring** is the graded ring

$$G(A) (= G_{\mathfrak{a}}(A)) = \bigoplus_{n=0}^{\infty} \mathfrak{a}^n / \mathfrak{a}^{n+1} \quad (a^0 = A).$$

Similarly, if M is an A -module and (M_n) is an $\mathfrak{a}$ -filtration of M , then the **associated graded module** is

$$G(M) = \bigoplus_{n=0}^{\infty} M_n / M_{n+1},$$

which is a graded $G(A)$ -module. Let $G_n(M) = M_n / M_{n+1}$.

Proposition 10.22. *Let A be a Noetherian ring, $\mathfrak{a}$ an ideal of A . Then*

- (i) $G_{\mathfrak{a}}(A)$ is Noetherian;
- (ii) $G_{\mathfrak{a}}(A)$ and $G_{\hat{\mathfrak{a}}}(\hat{A})$ are isomorphic as graded rings;
- (iii) *If M is a finitely-generated A -module and (M_n) is a stable $\mathfrak{a}$ -filtration of M , then $G(M)$ is a finitely-generated graded $G_{\mathfrak{a}}(A)$ -module.*

Proposition 10.23. *Let $\phi : A \rightarrow B$ be a homomorphism of filtered groups, i.e. $\phi(A_n) \subseteq B_n$, and let $G(\phi) : G(A) \rightarrow G(B)$, $\hat{\phi} : \hat{A} \rightarrow \hat{B}$ be the induced homomorphisms of the associated graded and completed groups. Then*

- (i) $G(\phi)$ injective $\Rightarrow \hat{\phi}$ injective;
- (ii) $G(\phi)$ surjective $\Rightarrow \hat{\phi}$ surjective.

Proposition 10.26. *If A is a Noetherian ring, $\mathfrak{a}$ an ideal of A , then the $\mathfrak{a}$ -adic completion $\hat{A}$ of A is Noetherian.*

DEFINITIONS AND PROPOSITIONS FROM EXERCISES

A Noetherian topological ring in which the topology is defined by an ideal contained in the Jacobson radical is called a **Zariski ring**.

Proposition (Hensel's lemma). *Let A be a local ring, $\mathfrak{m}$ its maximal ideal. Assume that A is $\mathfrak{m}$ -adically complete. If $f \in A[x]$ is monic of degree n , and if there exist coprime monic polynomials $\bar{g}, \bar{h} \in (A/\mathfrak{m})[x]$ of degrees $r, n-r$, with $\bar{f} = \bar{g}\bar{h}$, then we can lift $\bar{g}, \bar{h}$ back to monic polynomials $g, h \in A[x]$ such that $f = gh$.*

11 Dimension Theory

HILBERT FUNCTIONS

Let A be a Noetherian graded ring and M be a finitely-generated graded A -module. Let λ be an additive function on the class of all finitely generated A_0 -modules. The **Poincaré series** of M is the power series

$$P(M, t) = \sum_{n=0}^{\infty} \lambda(M_n) t^n \in \mathbb{Z}[[t]].$$

Theorem 11.1 (Hilbert-Serre theorem). $P(M, t)$ is a rational function in t of the form $f(t) / \prod_{i=1}^d (1 - t^{k_i})$, where $f(t) \in \mathbb{Z}[[t]]$.

The order of the pole of $P(M, t)$ at $t = 1$ is denoted as $d(M)$.

Corollary 11.2. If each $k_i = 1$, then for all sufficient large n , $\lambda(M_n)$ is a polynomial in n . More specifically, if $f(t) = \sum_{k=0}^N a_k t^k$, we have

$$\lambda(M_n) = \sum_{k=0}^n a_k \binom{d + n - k - 1}{d - 1}.$$

This is called **Hilbert function** or **Hilbert polynomial** of M , and its leading term is $(\sum a_k) n^{d-1} / (d-1)!$.

Proposition 11.3. If $x \in A_k$ is not a zero-divisor in M then $d(M/xM) = d(M) - 1$.

Proposition 11.4. Let A be a Noetherian local ring, $\mathfrak{m}$ its maximal ideal, $\mathfrak{q}$ an $\mathfrak{m}$ -primary ideal, M a finitely-generated A -module, (M_n) a stable $\mathfrak{q}$ -filtration of M . Then

- (i) M/M_n is of finite length for each $n \geq 0$;
- (ii) for all sufficiently large n this length is a polynomial $\chi_{\mathfrak{q}}^M(n)$ of degree $\leq s$, where s is the least number of generators of $\mathfrak{q}$;
- (iii) the degree and leading coefficient of $\chi_{\mathfrak{q}}^M(n)$ depend only on M and $\mathfrak{q}$, not on the filtration chosen.

If $M = A$, the polynomial $\chi_{\mathfrak{q}}^M(n)$ is called the **characteristic polynomial** of the $\mathfrak{m}$ -primary ideal $\mathfrak{q}$, simply noted as $\chi_{\mathfrak{q}}(n)$.

Proposition 11.6. $\deg \chi_{\mathfrak{q}}(n) = \deg \chi_{\mathfrak{m}}(n)$.

DIMENSION THEORY OF NOETHERIAN LOCAL RINGS

Theorem 11.14 (Dimension theorem). For any Noetherian local ring A the following three integers are equal:

- (i) the maximum length of chains of prime ideals in A (**Krull dimension**);
- (ii) the degree of the characteristic polynomial $\chi_{\mathfrak{m}} = l(A/\mathfrak{m}^n)$;
- (iii) the least number of generators of an $\mathfrak{m}$ -primary ideal of A .

Corollary 11.17 (Krull's principal ideal theorem). Let A be a Noetherian ring and let x be an element of A which is neither a zero-divisor nor a unit. Then every minimal prime ideal $\mathfrak{p}$ of (x) has height 1.

Corollary 11.18. Let A be a Noetherian local ring, x an element of $\mathfrak{m}$ which is not a zero-divisor. Then $\dim A/(x) = \dim A - 1$.

REGULAR LOCAL RINGS

Theorem 11.22. *Let A be a Noetherian local ring of dimension d , $\mathfrak{m}$ its maximal ideal, $k = A/\mathfrak{m}$. Then the following are equivalent:*

- (i) $G_{\mathfrak{m}}(A) \cong k[t_1, \dots, t_d]$ where the t_i are independent indeterminates;
- (ii) $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = d$;
- (iii) $\mathfrak{m}$ can be generated by d elements.

The ring satisfying any of the conditions is called *regular* local ring.

Proposition 11.23. *Let A be a ring, $\mathfrak{a}$ an ideal of A such that $\bigcap_n \mathfrak{a}^n = 0$. Suppose that $G_{\mathfrak{a}}(A)$ is an integral domain. Then A is an integral domain.*

Proposition 11.24. *Let A be a Noetherian local ring. Then A is regular iff $\hat{A}$ is regular.*

TRANSCENDENTAL DIMENSION

Assume that k is an algebraically closed field and let V be an irreducible affine variety over k . Let the coordinate ring of V be $A(V) = k[x_1, \dots, x_n]/I(V)$. Then $k(V) = \text{Frac } A(V)$ is a finitely generated extension of k and so has a finite transcendence degree over k , which is defined to be the *dimension* of V .

If P is a point with maximal ideal $\mathfrak{m}$ we call the Krull dimension $\dim A(V)_{\mathfrak{m}}$ the *local dimension* of V at P .

Theorem 11.25. *For any irreducible variety V over k the local dimension at any point is equal to its dimension.*

EXERCISES

1 Rings and Ideals

1.1. Consider $(1+x)(1-x+x^2-\cdots+(-x)^{n-1})$.

1.2. (i) Passing to $(A/\mathfrak{p})[x]$. $A/\mathfrak{p}$ is an integral domain, so $\deg(\overline{f\bar{g}}) = \deg(\overline{f}) + \deg(\overline{g})$. $\overline{f\bar{g}} = 1$ means $a_1, \dots, a_n \in \mathfrak{p}$. Use [Proposition 1.8](#).

(ii) a, b are nilpotents $\Rightarrow a+b$ is nilpotent. Note that $f-a_0 = xf_1$ is nilpotent.

(iii) Let $gf=0$ and g is of minimal degree. The leading coefficient $a_nb_m=0$, so a_ng has less degree than g and so $a_ng=0$. Same process to the second leading and following coefficients, we have $a_i g=0$ for all i .

(iv) (Gauss's lemma) Let $\mathfrak{a}, \mathfrak{b}$ and $\mathfrak{c}$ are generated by the coefficients of f, g and fg , respectively. We need to proof $\mathfrak{c} = (1) \Leftrightarrow \mathfrak{a}\mathfrak{b} = (1)$. $\mathfrak{c} \subseteq \mathfrak{a}\mathfrak{b}$ by the form of coefficients in fg . If $\mathfrak{c} \subsetneq \mathfrak{a}\mathfrak{b} = (1)$, let $\mathfrak{m}$ be the maximal ideal containing $\mathfrak{c}$, then $(A/\mathfrak{m})[x]$ is an integer domain. $\overline{f\bar{g}} = 0$ but $\overline{f}, \overline{g} \neq 0$.

1.3. Induction on the indeterminates.

1.4. Pick the y in [Proposition 1.9](#) to be the indeterminate, then use [Exercise 1.2\(i\)](#).

1.5. (i) $\Leftarrow$: $fg = \sum_{i=0}^{\infty} \left(\sum_{j=0}^i a_j b_{i-j} \right) x^i$, so we can find g such that $\sum_{j=0}^i a_j b_{i-j} = 0$.

(ii) Similar to [Exercise 1.2\(ii\)](#). Construct a counterexample where a_n have increasing nilpotencies.

(iii) Pick the y in [Proposition 1.9](#) to be 1. Use (i) on $1-f \Leftrightarrow 1-a_0$.

(iv),(v) $A \cong A[[x]]/(x)$. Use [Proposition 1.1](#).

1.6. $\mathfrak{J}$ is an ideal, so there is an idempotent $e \in \mathfrak{J}$ but $e \notin \mathfrak{R}$. Pick the y in [Proposition 1.9](#) to be 1.

1.7. Note that $x(1-x^{n-1}) = 0 \in \mathfrak{p}$.

1.8. Zorn's lemma but define the partial order reversely.

1.9. $\Rightarrow$: Apply [Proposition 1.8](#) to $A/\mathfrak{a}$.

$\Leftarrow$: Use [Proposition 1.11\(ii\)](#).

1.10. Use [Corollary 1.5](#) and [Corollary 1.6\(i\)](#)

1.11. (i) $x+x = (x+x)^2 = x^2+2x+x^2 = x+2x+x$.

(ii) Proof that every element in $A/\mathfrak{p}$ is 1.

(iii) $x+y-xy$.

1.12. Use [Corollary 1.5](#).

1.13. **1 is a monic polynomial but it is neither irreducible nor reducible.** A is an integral domain, so $\deg(fg) = \deg(f) + \deg(g)$. If $\mathfrak{a} = (1)$, apply the equation to $1 = \sum_f y_f f(x_f)$.

1.14. If $xy \in \mathfrak{m}$, xy is zero-divisor, so are x and y . By maximality, $\mathfrak{m} \subseteq \mathfrak{m} + (x) = \mathfrak{m}$.

1.15. (i) Apply [Proposition 1.8](#) to $A/\mathfrak{a}$ to proof the second equation.

(iv) Use $\sqrt{\mathfrak{a} \cap \mathfrak{b}} = \sqrt{\mathfrak{a}\mathfrak{b}}$.

1.16. Irreducible polynomials relate to prime ideals.

1.17. (i)-(iv) Use [Exercise 1.15](#).

(v) Let $(X_{f_i})_{i \in I}$ be an open cover. By [Exercise 1.15](#), $V(\{f_i\}_{i \in I}) = \emptyset$, so $\{f_i\}_{i \in I}$ generates the unit ideal. Let $1 = \sum_{i \in J} g_i f_i$ where J is a finite subset of I . If $\mathfrak{p} \notin \bigcup_{i \in J} X_{f_i}$, then $f_i \in \mathfrak{p}$ for all $i \in J$.

EXERCISES

(vi) Similar to (v), where the unit ideal shall change to the radical $\sqrt{(f)}$.

1.18. (iv) If $V(\mathfrak{p}_x) \subseteq V(\mathfrak{p}_y)$, $\mathfrak{p}_x \in \overline{V(\mathfrak{p}_y)}$. If they don't have containing relation, $\mathfrak{p}_x \in \overline{V(\mathfrak{p}_y)}$ and $\mathfrak{p}_y \in \overline{V(\mathfrak{p}_x)}$.

1.19. Use the closed sets definition. Consider minimal prime ideals.

1.20. (i) Let $\overline{Y} = Z_1 \cup Z_2$, then $Z_i \cap Y = \emptyset$ for an $i \in \{1, 2\}$. Thus $Z_i = \overline{Z_i \cap Y} = \emptyset$.

(ii) Similar to (i). Construct the maximal element and proof that it is irreducible.

(iii) If Y Hausdorff but irreducible, every pair of open sets in Y intersect. Thus Y is singleton.

(iv) Use the closed set definition.

1.21. Simplify the sets: $\phi^{*-1}(E) = \{\mathfrak{q} \in Y : \phi^{-1}(\mathfrak{q}) \in E\}$, $\phi^*(V(E)) = \{\phi^{-1}(\mathfrak{q}) : E \subseteq \mathfrak{q}\}$.

1.22. Use quotient to find the form of prime ideals in A .

1.23. (i) $X_f = V(1 - f)$.

(ii) $X_{f_1} \cup X_{f_2} = X_{f_1 + f_2 - f_1 f_2}$.

(iii) A closed set in compact space is compact. By [Exercise 1.17](#), it is a finite union of basic open set. Then apply (ii).

(iv) By [Exercise 1.11](#) every prime ideal is maximal. Pick $f \in \mathfrak{p}_x \setminus \mathfrak{p}_y$, then X_f and $V(f)$ separate two points.

1.24. $a = a \wedge 1 = a \wedge (a \vee a') = (a \wedge a) \vee (a \wedge a') = (a \wedge a) \vee 0 = a \wedge a$.

1.25. Use [Exercise 1.23\(iv\)](#).

1.26. $\mathfrak{m}_x \in \mu(U_f) \Leftrightarrow f(x) \neq 0 \Leftrightarrow f \notin \mathfrak{m}_x \Leftrightarrow \mathfrak{m}_x \in \tilde{U}_f$.

1.27. If $x \neq y$, there exists a polynomial $f \in P(X)$ such that $f(x) = 0$ but $f(y) \neq 0$, so $\mathfrak{m}_x \neq \mathfrak{m}_y$, and so μ injective. By Nullstellensatz, every maximal ideal in $P(X)$ is of the form $\mathfrak{m} = (t_1 - a_1, \dots, t_n - a_n)$, with $(a_1, \dots, a_n) \in X$, so μ surjective. By [Exercise 1.26](#), μ is a homeomorphism.

1.28. Project to coordinate functions.

2 Modules

2.1. Bézout's identity.

2.2. Tensor the exact sequence $0 \rightarrow \mathfrak{a} \rightarrow A \rightarrow A/\mathfrak{a} \rightarrow 0$. Calculate the image of $\mathfrak{a} \otimes M \rightarrow A \otimes M$.

2.3. Consider the residue field k of A and pass $M \otimes_A N$ to vector space $M_k \otimes_k N_k$. Use [Exercise 2.2](#) and [Nakayama's lemma](#).

2.4. There exist canonical projections.

2.5. The definition of flatness only considers the module structure. Use [Exercise 2.4](#).

2.6. Construct two homomorphisms.

2.7. Proof that $(A/\mathfrak{p})[x]$ and $A[x]/\mathfrak{p}[x]$ are isomorphic.

2.8. (ii) Use [Proposition 2.15](#).

2.9. Pick $y_1, \dots, y_m \in M$ whose images are the generators of M'' . Use diagram chasing for an element of M .

2.10. $u(m) + \mathfrak{a}N \in N$, then use surjectiveness. Apply [Nakayama's lemma](#) on $(u(M) + \mathfrak{a}N)/N$.

2.11. (i),(ii) Consider a residue field and pass to vector spaces. Use the right exactness of tensor product.

(iii) If $m > n$, let $\psi : A^m \xrightarrow{\phi} A^n \hookrightarrow A^m$. By [Proposition 2.4](#), ψ satisfies $\psi^n + a_{n-1}\psi^{n-1} + \cdots + a_0 = 0$. Pick such equation to have the minimum degree, then ψ injective implies $a_0 \neq 0$. Apply the equation to element $(0, \dots, 0, 1)$ to obtain contradiction.

2.12. Choose $u_i \in M$ such that $\phi(u_i)$ are a basis of A^n , then proof that $0 \rightarrow \ker(\phi) \rightarrow M \xrightarrow{\phi} A^n \rightarrow 0$ splits.

2.13. $p \circ g = \text{id}_N$ injective $\Rightarrow g$ injective, so $0 \rightarrow N \xrightarrow{g} N_B \rightarrow N_B/g(N) \rightarrow 0$ splits.

2.14. This question asks to proof $\mu_i = \mu_j \circ \mu_{ij}$.

2.15. (i) Use [Exercise 2.14](#) to increase indices.

(ii) $x_i = \sum_k (y_k - \mu_{ki}(y_k))$ is an equation in $\bigoplus M_i$, so μ_{ij} cannot be applied directly. Use canonical projection $\pi_i : \bigoplus M_i \rightarrow M_i$ on the equation first.

2.16 (Universal property of direct limit). Use [Exercise 2.15\(i\)](#) for both existence and uniqueness.

2.17. Similar to [Exercise 2.15\(i\)](#), move components into a same index.

2.18. Use [Exercise 2.16](#).

2.19 ($\varinjlim$ is an exact functor). Diagram chasing. Use [Exercise 2.15](#).

2.20. Draw the diagram. Define two homomorphisms by the universal properties of tensor product and direct limit. Verify their compositions are identity maps.

2.21. Use [Exercise 2.15\(ii\)](#) on $\alpha_i(1)$.

2.22. Use [Exercise 2.15](#). Pick a nilpotent or assume $xy = 0$.

2.23. If $J \subseteq J'$, $B_J \cong B_J \otimes (A)_{i \in J' \setminus J}$. Let Γ be the set of all finite subsets of Λ , then $(B_J)_{J \in \Gamma}$ is the case of [Exercise 2.21](#). We can also use [Exercise 2.17](#) to see its form.

2.24. Directly from the definition of Tor functor.

2.25. Use [Snake lemma](#).

2.26. Let M_n be generated by $\{x_1, \dots, x_n\}$. We have an exact sequence

$$\text{Tor}_1(M_{n-1}, N) \rightarrow \text{Tor}_1(M_n, N) \rightarrow \text{Tor}_1(M_n/M_{n-1}, N).$$

By [Proposition 2.3](#), $M_n/M_{n-1} \cong A/\mathfrak{a}$.

Let $f : \mathfrak{a} \otimes N \rightarrow A \otimes N$, $f(\sum a_i \otimes y_i) = 0$. Let $\mathfrak{a}'$ be generated by a_i . By assumption, $f|_{\mathfrak{a}' \otimes N}$ is injective, so $\sum a_i \otimes y_i = 0$ and thus $\text{Tor}_1(A/\mathfrak{a}, N) = 0$. Since $M_0 = A$ and $\text{Tor}_1(A, N) = 0$, the exact sequence implies $\text{Tor}_1(M_n, N) = 0$ for all n . Then use [Proposition 2.19\(iv\)](#).

Another approach is to use the fact that any module is the direct limit of some finitely generated modules, and direct limit is an exact functor.

2.27. (i) $\Rightarrow$ (ii) $A/(x)$ is a flat A -module, then $\phi : A/(x) \otimes (x) \rightarrow A/(x)$ injective. $\phi(1 \otimes x) = x = 0$, so $A/(x) \otimes (x) = 0$. Then use [Exercise 2.2](#).

(ii) $\Rightarrow$ (iii) Let $x = ax^2$, then ax is an idempotent. $(x) \ni bx = bax^2 = (bx)(ax) \in (ax)$, so $(ax) = (x)$. Similar to [Exercise 1.11\(iii\)](#), every finitely generated ideal is generated by an idempotent e , so $A = (e) \oplus (1 - e)$.

(iii) $\Rightarrow$ (i) $N \otimes_A A = (N \otimes_A \mathfrak{a}) \oplus (N \otimes_A \mathfrak{a}')$, so $(N \otimes_A \mathfrak{a}) \rightarrow N \otimes_A A$ injective. Use [Exercise 2.26](#).

2.28. Use [Exercise 2.27\(ii\)](#).

3 Rings and Modules of Fractions

3.1. The only if part requires the finitely generated property.

3.2. Use [Proposition 1.9](#). Construct the inverse by elementary algebra.

If $M = \mathfrak{a}M$, $S^{-1}M = S^{-1}\mathfrak{a}$, $S^{-1}M = (S^{-1}\mathfrak{a})(S^{-1}M)$. Then use [Nakayama's lemma](#) and [Exercise 3.1](#).

3.3. Construct the isomorphism by $(a/s)/(t/1) \mapsto a/(st)$.

3.4. Construct the isomorphism.

3.5. If $a \in A$ is a nilpotent, let $\mathfrak{p} \supseteq \text{Ann}(a)$. $a/1 \in A_{\mathfrak{p}}$ is zero, so $sa = 0$ for some $s \in A - \mathfrak{p}$.

Counter example is $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

3.6. Use the fact that for any multiplicatively closed set S , if $A - S$ is an ideal, it's automatically prime.

3.7. (i),(ii) More specifically, $A - \overline{S} = \bigcup_{\mathfrak{p} \cap S = \emptyset} \mathfrak{p}$.

(iii) $\mathfrak{p} \cap (1 + \mathfrak{a}) = \emptyset \Leftrightarrow \mathfrak{a} + \mathfrak{p} \neq (1) \Leftrightarrow \mathfrak{a} + \mathfrak{p} \subseteq \mathfrak{m}$ for some maximal ideal $\mathfrak{m}$. Note that the union of a chain of prime ideals is only a maximum ideal.

3.8. (v) $\Rightarrow$ (ii) If $t/1$ is not a unit in $S^{-1}A$, it is contained in a prime ideal $\mathfrak{q} \subseteq S^{-1}A$, so $t \in T \cap \mathfrak{q}^c$.

3.9. Let S be maximal in [Exercise 3.6](#). If $S_0 \not\subseteq S$, $S \subsetneq SS_0$. Then use [Exercise 3.6](#).

(i) Assume that S_1 is larger, then pick a zero-divisor $x \in S_1$.

(ii) If a/s is not a zero-divisor, a is not a zero-divisor, so $a \in S_0$.

(iii) Note that $s \in S_0$ is a unit.

3.10. (i) [Exercise 2.27](#) says for any element $x \in A$, $x = ax^2$ for some $a \in A$.

(ii) $\Rightarrow$: Use [Exercise 2.28](#). $\Leftarrow$: Consider an $A_{\mathfrak{m}}$ -module. Use [Proposition 3.10](#).

3.11. (i) $\Rightarrow$ (ii) Use [Exercise 2.27](#) on $A/\mathfrak{p}$.

(ii) $\Rightarrow$ (i) Proof that $(A/\mathfrak{A})_{\overline{\mathfrak{p}}}$ has no nilpotent, then use [Exercise 1.10](#) and [Exercise 3.10](#).

(ii) $\Rightarrow$ (iv) Let $f \in \mathfrak{p}_1$ but $f \notin \mathfrak{p}_2$, then $f/1 \in A_{\mathfrak{p}_1}$ is nilpotent, so $sf^n = 0$. We have $\mathfrak{p}_1 \in X_s$ and $\mathfrak{p}_2 \in X_{f^n}$. By [Exercise 1.17](#), $X_s \cap X_{f^n} = X_{sf^n} = \emptyset$.

3.12. (i)-(iii) Use the torsion element definition.

(iv) Use [Proposition 3.5](#).

3.13. Proof they contain in each other. Use [Proposition 3.8](#) for the equivalent conditions.

3.14. Pass to $M/\mathfrak{a}M$. Use [Proposition 3.8](#).

3.15. Let $\{x_i\}_{i=1}^n$ be generators and $\{e_i\}_{i=1}^n$ be the canonical basis. Let $\phi(e_i) = x_i$, so ϕ is surjective.

Consider a residue field $k = A/\mathfrak{m}$, we have $\text{Tor}_1(k, F) \rightarrow k \otimes \ker \phi \rightarrow k \otimes F \xrightarrow{1 \otimes \phi} k \otimes F \rightarrow 0$. $\text{Tor}_1(k, F) = 0$ by balancing Tor theorem; $1 \otimes \phi$ bijective so $k \otimes \ker \phi = 0$. By [Exercise 2.3](#), $\ker \phi = 0$.

If $\{x_i\}_{i=1}^m$ generate F where $m < n$, pick some element $\{x_i\}_{i=m+1}^n$ so we also have $\phi(e_i) = x_i$. By injectiveness of ϕ , $\{e_i\}_{i=1}^n$ are linearly dependent.

3.16. (i) $\Rightarrow$ (ii) $f^*(p^e) = p$.

(ii) $\Rightarrow$ (iii) Use [Proposition 1.17](#).

(iii) $\Rightarrow$ (iv) Every module contains a cyclic submodule, which is isomorphic to $A/\mathfrak{a}$. Then use $B \otimes_A A/\mathfrak{a} \cong B/\mathfrak{a}^e$ and $\mathfrak{a}^e \subseteq \mathfrak{m}^e$.

(iv) $\Rightarrow$ (v) Let the mapping be ϕ . Since B flat, $0 \rightarrow (\ker \phi)_B \rightarrow M_B \rightarrow (M_B)_B$ exact. By [Exercise 2.13](#), $M_B \rightarrow (M_B)_B$ injective, so $(\ker \phi)_B = 0$; by assumption, $\ker \phi = 0$.

(v) $\Rightarrow$ (i) Take $M = A/\mathfrak{a}$. Then $M_B = B/\mathfrak{a}^e$.

3.17. If $M \rightarrow N$ injective, we have a diagram

$$\begin{array}{ccc} B \otimes_A M & \xrightarrow{\phi} & B \otimes_A N \\ \downarrow g \otimes \text{id}_M & & \downarrow g \otimes \text{id}_N \\ C \otimes_A M & \xrightarrow{\psi} & C \otimes_A N \end{array}$$

$g \circ f$ flat, so ψ injective. g faithfully flat, by [Exercise 3.16\(v\)](#), $B \otimes_A M \rightarrow C \otimes_B (B \otimes_A M)$ injective. By [Proposition 2.15](#), $C \otimes_B (B \otimes_A M) = C \otimes_A M$, so $g \otimes \text{id}_M$ injective.

3.18. Use [Exercise 3.3](#) to proof $B_{\mathfrak{q}} \cong (B_{\mathfrak{p}})_{\mathfrak{q}B_{\mathfrak{p}}}$. By [Proposition 3.3](#) and [Proposition 3.5](#), $B_{\mathfrak{q}}$ is flat over $B_{\mathfrak{p}}$. Hence $B_{\mathfrak{q}}$ is flat over $A_{\mathfrak{p}}$ and satisfies [Exercise 3.16\(iii\)](#).

3.19. (v) $M \cong \bigoplus_{i=1}^n A/\mathfrak{a}_i$, so $\text{Supp}(M) = \bigcup_{i=1}^n \text{Supp}(A/\mathfrak{a}_i) = \bigcup_{i=1}^n V(\mathfrak{a}_i) = V(\bigcap_{i=1}^n \mathfrak{a}_i)$. Proof that $\bigcap_{i=1}^n \mathfrak{a}_i = \text{Ann}(M)$.

(vi) Use [Exercise 2.3](#).

(vii) Use [Exercise 2.2](#).

(viii) $\mathfrak{q} \in f^{*-1}(\text{Supp}(M)) \Leftrightarrow M_{\mathfrak{q}^c} \neq 0 \Leftrightarrow M_{\mathfrak{p}} \neq 0$ and $B_{\mathfrak{q}} \neq 0 \Leftrightarrow \mathfrak{q} \in \text{Supp}(M) \cap \text{Supp}(B)$.

3.20. Counter example is $A[x]/(x^2)$.

3.21. (i),(ii) Use [Proposition 3.11\(iv\)](#).

(iii) Use [Exercise 1.21\(iv\)](#).

(iv) To proof $f^{*-1}(\mathfrak{p}) = \text{Spec}(B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}})$, use $\mathfrak{q} \in \text{Spec}(S^{-1}B) \Leftrightarrow \mathfrak{q} \cap f(S) = \emptyset$ and $\mathfrak{q} \in \text{Spec}(C/\mathfrak{p}C) \Leftrightarrow f^{-1}(\mathfrak{q}) \supseteq \mathfrak{p}$. To proof the equation, note that $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} = \text{Frac}(A/\mathfrak{p})$, then apply [Proposition 2.15](#) on $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \otimes_{A/\mathfrak{p}} A/\mathfrak{p} \otimes_A B$.

3.22. Let $\mathfrak{p} \in X_f$, so $f \notin \mathfrak{p}$. Then $\mathfrak{q} \in \text{Spec}(A_{\mathfrak{p}}) \Leftrightarrow \mathfrak{q} \subseteq \mathfrak{p} \Leftrightarrow f \notin \mathfrak{q} \Leftrightarrow \mathfrak{q} \in X_f$.

3.23. (i) Use [Exercise 1.17\(iv\)](#).

(v) Define two homomorphisms by the universal properties of direct limit and local ring.

3.24.

3.25.

3.26.

3.27.

3.28.

3.29.

3.30.

4 Primary Decomposition

4.1. Use [Exercise 1.21\(iv\)](#), so $\text{Spec}(A/\mathfrak{a}) = V(\mathfrak{a})$.

4.2. All primary components are prime.

4.3. For any element $x \in A$, $x(1 - ax) = 0 \in \mathfrak{q}$.

4.4. $\mathfrak{m} = \sqrt{\mathfrak{q}} \supsetneq \mathfrak{q} \supsetneq \mathfrak{m}^2$.

4.5. $\mathfrak{m}^2$ is $\mathfrak{m}$ -primary.

4.6.

4.7. (ii),(iv),(v) $(\mathfrak{a}[x])^c = \mathfrak{a}$.

EXERCISES

(iii) Let $\bar{f}, \bar{g} \in (A/\mathfrak{q})[x]$. If $\bar{f}\bar{g} = 0$ but $\bar{g} \neq 0$, by [Exercise 1.2\(iii\)](#), $\bar{a}\bar{f} = 0$, so $aa_i \in \mathfrak{q}$. Hence $a_i \in \mathfrak{p}$ and $a_i^{n_i} \in \mathfrak{q}$, so $\bar{a}_i^{n_i} = 0$ and use [Exercise 1.2\(ii\)](#).

4.8. $k[x_1, \dots, x_n]/\mathfrak{p}_i = k[x_{i+1}, \dots, x_n]$ integer domain, so $\mathfrak{p}_i$ prime. Note that $\mathfrak{p}_i^n$ is the set of all homogeneous polynomials in $k[x_1, \dots, x_i]$ of degree $\geq n$.

4.9. (i) $\Leftarrow$: x is contained in all prime ideals containing $(0 : a)$, so $x \in \sqrt{(0 : a)}$.

(ii) If $\mathfrak{a} \subseteq \mathfrak{p}$ but $\mathfrak{a}^e \subseteq S^{-1}\mathfrak{p}' \subsetneq S^{-1}\mathfrak{p}$, then $\mathfrak{a} \subseteq \mathfrak{a}^{ee} \subseteq \mathfrak{p}' \subsetneq \mathfrak{p}$. If $\mathfrak{b} \subseteq S^{-1}\mathfrak{p}$ but $\mathfrak{b}^e \subseteq \mathfrak{p}' \subsetneq \mathfrak{p}$, use $\mathfrak{b}^{ee} = \mathfrak{b}$ derived from [Proposition 3.11\(i\)](#). $S^{-1}\mathfrak{p}$ is a prime ideal if S doesn't meet $\mathfrak{p}$.

(iii) $(0 : a) = \bigcap (\mathfrak{q}_i : a) \subseteq \mathfrak{p}$. By [Proposition 1.11\(ii\)](#), $(\mathfrak{q}_i : a) \subseteq \mathfrak{p}$ for some i . Then use [Proposition 4.4](#) and the minimality of $\mathfrak{p}$.

4.10. (iv) Let $x \neq 0$, $(0 : x) \subseteq \mathfrak{p}$ for some $\mathfrak{p}$. If $x \in S_{\mathfrak{p}}(0)$, then $sx = 0$, so $s \in (0 : x)$.

4.11. Note that $x \in S(\mathfrak{a}) \Leftrightarrow$ there exists an $s \in S$ such that $sx \in \mathfrak{a}$. Show that $S_{\mathfrak{p}}(0)$ is contained in all $\mathfrak{p}$ -primary ideals.

If $0 = \bigcap \mathfrak{q}_i$ is irredundant, $\sqrt{(0)}$ is the intersection of minimal prime ideals.

4.12. Use [Proposition 3.11\(ii\)](#). Use [Proposition 4.9](#) to show there are at most 2^n different $S(\mathfrak{a})$.

4.13. (i) Use [Exercise 4.12](#).

(ii) Use [Proposition 4.8](#) and [Proposition 4.9](#).

(iii) Proof that $S_{\mathfrak{p}}(\mathfrak{p}^{(m)}\mathfrak{p}^{(n)}) = \mathfrak{p}^{(m+n)}$.

(iv) $\Leftarrow$: Show that $\mathfrak{p}^{(n)}$ is the smallest $\mathfrak{p}$ -primary ideal containing $\mathfrak{p}^n$.

4.14. Let $x \notin \mathfrak{q}$. If $ab \in (\mathfrak{q} : x)$ but $a \notin (\mathfrak{q} : x)$, we have $xab \in \mathfrak{q}$ but $xa \notin \mathfrak{q}$. Thus, $b \in (\mathfrak{q} : xa) = (\mathfrak{q} : x)$ by maximality, and therefore $(\mathfrak{q} : x)$ is prime.

4.15. Use [Proposition 4.9](#).

4.16. Use [Proposition 3.11\(iv\)](#) and [Proposition 4.9](#).

4.17.

4.18. (i) $\Rightarrow$ (L1) By [Proposition 4.9](#), $S_{\mathfrak{p}}(\mathfrak{a}) = \bigcap \mathfrak{q}_i$. Pick $x \in S_{\mathfrak{p}}$, then $\mathfrak{q}_i = (\mathfrak{q}_i : x)$ by [Proposition 4.4](#).

(i) $\Rightarrow$ (L2) Use [Exercise 4.12](#).

4.19. Similar to the proof of [Exercise 4.11](#).

The goal is to prove that there exists $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ irredundant, so we need to prove $\mathfrak{q}_i \not\supseteq \bigcap_{j \neq i} \mathfrak{q}_j$. Using the fact that $u\mathfrak{v} \subseteq \mathfrak{q}$ but $u \not\subseteq \mathfrak{q} \Rightarrow \mathfrak{v} \subseteq \mathfrak{p}$, we can assume $\mathfrak{b} = \bigcap_{i=1}^{n-1} \mathfrak{q}_i$ irredundant and $\mathfrak{p}_n$ maximal among $\{\mathfrak{p}_i\}$, so we only need to find $\mathfrak{q}_n \not\supseteq \mathfrak{b}$. If such $\mathfrak{q}_n$ doesn't exist, $\mathfrak{b} \subseteq S_{\mathfrak{p}_n}(0)$. Pick a minimal prime ideal $\mathfrak{p} \subseteq \mathfrak{p}_n$ and use [Exercise 4.10](#) and [Proposition 1.11\(ii\)](#).

4.20.

4.21.

4.22.

4.23.

5 Integral Dependence and Valuations

5.1. $f^*(V(I)) = \{f^{-1}(\mathfrak{q}) : \mathfrak{q} \supseteq I\}$. Use [Theorem 5.10](#) to prove $f^*(V(I)) \supseteq V(f^{-1}(I))$.

5.2. $\mathfrak{p} = \ker f$ is prime. By [Theorem 5.10](#), there is a prime ideal $\mathfrak{q} \subseteq B$. $\text{Frac}(B/\mathfrak{q})$ is an algebraic extension of $\text{Frac}(A/\mathfrak{p})$, so $\text{Frac}(A/\mathfrak{p}) \rightarrow \Omega$ can be extended to $\text{Frac}(B/\mathfrak{q}) \rightarrow \Omega$.

5.3. Use the operator $- \otimes_A c^n$.

5.4. If $\left(\frac{1}{x+1}\right)^n + \sum \frac{f_i(x^2-1)}{g_i(x^2-1)(x+1)^i} = 0$, multiply both side by $(x+1)^n$ and take $x = -1$ to get contradiction.

5.5. (ii) Use [Theorem 5.10](#) to proof that the contraction of maximal ideals in B are maximal in A .

5.6. Let $b_i \in B_i$, $f_i \in A[x_i]$ monic, $f_i(b_i) = 0$. Let $f = (f_1, \dots, f_n)$, then $f(b_1, \dots, b_n) = 0$.

5.7. Let $x \in B$ and x integral over A . Let $x^n + a_{n-1}x^{n-1} + \dots + a_0 = 0$ is the equation of the least degree. Analyze $x(x^{n-1} + a_{n-2}x^{n-2} + \dots + a_1) = -a_0$.

5.8. Proof that there is a ring $D \supseteq B$ such that f and g factor into linear factors in D . Use Kronecker's construction.

5.9. Let $f^n + g_{n-1}f^{n-1} + \dots + g_0 = 0$. Let $f_1 = f - x^r$ with r larger than all $\deg g_i$, then $f_1(f_1^{n-1} + h_{n-1}f_1^{n-2} + \dots + h_1) = -h_0$ with h_0 monic. Then use [Exercise 5.8](#).

5.10. (a) $\Rightarrow$ (b) Let $f^*(V(\mathfrak{q}_1)) = V(I)$, so $I \subseteq \mathfrak{p}_1 \subseteq \mathfrak{p}_2$, and we have $\mathfrak{p}_2 \in V(I) = f^*(V(\mathfrak{q}_1))$.
(a') $\Rightarrow$ (c')

5.11. Use [Exercise 3.18](#) and [Exercise 5.10](#).

5.12. Note that G has identity element so x is a root of the polynomial $\prod_{\sigma \in G} (t - \sigma(x))$. The coefficients are symmetric polynomials of σ_i . **G acting on $S^{-1}A$ is defined to be $\sigma(x/s) = \sigma(x)/\sigma(s)$.**

5.13. Let $\mathfrak{q} \in P$. Then $\sigma(\mathfrak{q}) \cap A^G = \mathfrak{q} \cap A^G = \mathfrak{p}$, so $\sigma(\mathfrak{q}) \in P$. Let $\mathfrak{q}_1, \mathfrak{q}_2 \in P$ and $x \in \mathfrak{q}_1$. Then $\prod_{\sigma \in G} \sigma(x) \in \mathfrak{q}_1 \cap A^G = \mathfrak{p} \subseteq \mathfrak{q}_2$, so $x \in \sigma^{-1}(\mathfrak{q}_2)$, and so $\mathfrak{q}_1 \subseteq \bigcup_{\sigma \in G} \sigma(\mathfrak{q}_2)$. Use [Proposition 1.11\(i\)](#) and [Corollary 5.9](#).

5.14. Note that $\sigma \in G$ fixes A .

5.15.

5.16.

5.17. Consider $k[t_1, \dots, t_n]/I(X)$. Use [Exercise 5.16](#).

5.18 (Zariski's lemma). Let n be the number of generators. If $n = 1$, x_1 is invertible, so $x_1^{-1} = \sum_{i=1}^n c_i x_1^i$.

If $n > 1$, let $A = k[x_1]$ and $K_A = \text{Frac}(A)$. Then $K_A[x_2, \dots, x_n]$ is a finite algebraic extension of K_A , so there are monic polynomials $f_i \in K_A[y]$ such that $f_i(x_i) = 0$ for $2 \leq i \leq n$. Pick $a \in A$ such that $a f_2 \dots f_n \in A[y]$, then $x_2, \dots, x_n$ are integral over A_a . Therefore, $K_A \subseteq B$ are integral over A_a .

If x_1 transcendental over k , A is a UFD. Since every UFD is integrally closed, A is integrally closed, so A_a is integrally closed by [Proposition 5.12](#). K_A integral over A_a , so $A_a = K_A$, but such a must be a unit in A .

5.19. By [Exercise 5.18](#), $k[t_1, \dots, t_n]/\mathfrak{m}$ is a finite algebraic extension of k , so $k[t_1, \dots, t_n]/\mathfrak{m} \cong k$. Let $t_i \mapsto a_i$, then $t_i - a_i \in \mathfrak{m}$. Proof that $(t_1 - a_1, \dots, t_n - a_n)$ is maximal.

5.20. Hint.

5.21. We have

$$\begin{array}{ccccccc} A & \xrightarrow{\phi_A} & A_s & \longrightarrow & A_s[y_1, \dots, y_n] & \longrightarrow & B_s \xleftarrow{\phi_B} B \\ & & \searrow f & & \searrow f_s & & \downarrow g \\ & & & & & & \Omega \end{array}$$

f_s by universal property of localization. f'_s by defining $y_i \mapsto 0$. By [Exercise 5.20](#), B_s integral over $A_s[y_1, \dots, y_n]$, so g is obtained by [Exercise 5.2](#).

5.22. Hint.

EXERCISES

5.23. (i) $\Rightarrow$ (ii) Note that $f : A \rightarrow A/\ker f$. Use [Proposition 1.1](#).

(ii) $\Rightarrow$ (iii) Consider $A/\mathfrak{p}$.

(iii) $\Rightarrow$ (i) If $\mathfrak{p}$ is not an intersection of maximal ideals, the Jacobson radical $\mathfrak{J}$ of $A/\mathfrak{p}$ is not zero. Pick $f \in \mathfrak{J}$, then $(A/\mathfrak{p})_f \neq 0$. The contraction of a maximal ideal of $(A/\mathfrak{p})_f$ is $\mathfrak{q} \in A$ which is not maximal and doesn't contain f , but all prime ideals strictly containing $\mathfrak{q}$ contains f .

5.24. (i) Use [Exercise 5.23\(iii\)](#) and [going-up theorem](#).

(ii) Consider $B/\mathfrak{q}$ and $A/\mathfrak{q}^c$. Then use [Exercise 5.22](#), [Exercise 5.23\(i\)](#) and [Proposition 1.1](#).

5.25. (i) $\Rightarrow$ (ii)

(ii) $\Rightarrow$ (i)

5.26. (2) $\Rightarrow$ (3) $U_1 \cap X_0 = U_2 \cap X_0 \Rightarrow U_1^c \cap X_0 = U_2^c \cap X_0$, then take closure.

(i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) Consider $V(\mathfrak{a}) \cap X_f \cap \text{Max}(A)$. If all maximal ideals in $V(\mathfrak{a})$ contains f , $V(\mathfrak{a}) \cap X_f = \emptyset$ by [Exercise 5.23\(iii\)](#).

(iii) $\Rightarrow$ (i) If not Jacobson, there is a prime ideal $\mathfrak{p}$ such that $\mathfrak{p} \subsetneq \bigcap_{\mathfrak{q} \supseteq \mathfrak{p}} \mathfrak{q}$. Let $f \in \bigcap_{\mathfrak{q} \supseteq \mathfrak{p}} \mathfrak{q} \setminus \mathfrak{p}$, then $X_f \cap V(\mathfrak{p}) = \{\mathfrak{p}\}$.

5.27. $\Rightarrow$: Use [Theorem 5.21](#) to proof maximal elements are valuation rings.

$\Leftarrow$: If $(A, \mathfrak{m}) \subsetneq (A', \mathfrak{m}')$, pick $x \in A' \setminus A$. Then $x^{-1} \in \mathfrak{m}$ because it is not a unit in A .

5.28. (1) $\Rightarrow$ (2) Pick $a \in \mathfrak{a} \setminus \mathfrak{b}$, and any $b \in \mathfrak{b}$. Consider a/b and b/a .

(2) $\Rightarrow$ (1) Consider $(a) \subseteq (b)$ or $(b) \subseteq (a)$, we have $a = xb$ or $b = ya$, so $y = x^{-1}$.

5.29. B is a local ring of A means $B = A_{\mathfrak{p}}$ for some prime ideal $\mathfrak{p}$. Let $\mathfrak{m}$ be the maximal ideal of B and let $\mathfrak{p} = \mathfrak{m} \cap A$. If $s \in A$ but $s \notin \mathfrak{p}$, then $s \in B$ and $s \notin \mathfrak{m}$, so s is a unit in B . Let $A_{\mathfrak{p}} \rightarrow B$ to be $x/s \mapsto s^{-1}x$, which is injective; $\mathfrak{p}A_{\mathfrak{p}} \subseteq B$, so B dominates $A_{\mathfrak{p}}$. By [Exercise 5.28](#), $A_{\mathfrak{p}}$ is a valuation ring, then use [Exercise 5.27](#).

5.30. $xy^{-1} \in A$ or $x^{-1}y \in A$, so $\geq$ is a total order. Also $(xz)(yz)^{-1} = xy^{-1}$.

Treat v as a representation map. If $v(x) \geq v(y)$, $(x+y)y^{-1} = xy^{-1} + 1 \in A$.

5.31. Use the given conditions to verify the set is closed under addition and multiplication. Then verify it is a valuation ring by taking $x = y = 1$ and $y = x^{-1}$ in the first condition.

5.32.

5.33.

5.34.

5.35.

6 Chain Conditions

6.1. (i) Let $x \in \ker(u)$. Since u^n surjective, there is a $y \in M$ such that $u^n(y) = x$. But $u^{n+1}(y) = u(x) = 0$, means that $y \in \ker(u^{n+1}) = \ker(u^n)$.

(ii) Let $x \in M$. Then $u^n(x) \in \text{im}(u^n) = \text{im}(u^{n+1})$, so there is a $y \in M$ such that $u^{n+1}(y) = u^n(x)$. But u^n injective, means that $u(y) = x$.

6.2. If M is not Noetherian, there is a submodule N which is not finitely generated. Construct a chain of finitely generated submodules by the generators of N .

6.3. Consider $0 \rightarrow N_2/(N_1 \cap N_2) \rightarrow M/(N_1 \cap N_2) \rightarrow M/N_2 \rightarrow 0$. Use [Proposition 2.1](#).

6.4. Let $m_1, \dots, m_n$ be generators of M , then $\mathfrak{a} = \bigcap \text{Ann}(m_i)$. $A/\text{Ann}(m_i) \cong Am_i \subseteq M$, then use [Exercise 6.3](#).

6.5. Use the open set definition.

6.6. (i) $\Rightarrow$ (iii) Use [Exercise 6.5](#).

(ii) $\Rightarrow$ (i) Consider a chain of ascending open sets in X .

6.7. Let Σ be the collection of closed subsets of X which are not finite union of irreducible closed subspaces. Then Σ has minimal element X_0 . Proof that X_0 cannot be either irreducible or reducible.

6.8. The converse isn't true. The counter example is $k[x_1, \dots]/(x_1^2, \dots)$.

6.9. The irreducible components of $\text{Spec}(A)$ is $V(\mathfrak{p})$ where $\mathfrak{p}$ minimal ([Exercise 1.20\(iv\)](#)).

6.10. Use [Exercise 3.19\(v\)](#).

6.11. By [Exercise 6.7](#), $V(\mathfrak{b})$ is a finite union of $V(\mathfrak{q}_i)$, where $\mathfrak{q}_i$ minimal and containing $\mathfrak{b}$. Then use [Exercise 5.10\(c\)](#).

6.12. The converse isn't true. The counter example is an infinite product $\prod \mathbb{Z}/2\mathbb{Z}$.

7 Noetherian Rings

7.1.

7.2.

7.3. (i) $\Rightarrow$ (ii) Use [Proposition 4.8](#) and [Proposition 4.4](#).

(ii) $\Rightarrow$ (iii) Let $S = \{1, x, x^2, \dots\}$. Proof that $\bigcup (\mathfrak{a} : x^n) = S(\mathfrak{a})$.

(iii) $\Rightarrow$ (i) Similar to the proof of [Proposition 7.12](#).

7.4.

7.5. Use [Exercise 5.12](#) and [Proposition 7.8](#).

7.6. Hint.

7.7. Use a.c.c..

7.8. $A \cong A[x]/(x)$.

7.9. Hint.

7.10. $M[x] = M \otimes A[x]$, so it is a homomorphic image of $M \times A[x]$.

7.11. Counter example is an infinite product $\prod \mathbb{Z}/2\mathbb{Z}$.

7.12. By [Exercise 3.16\(i\)](#), $\mathfrak{a} \mapsto \mathfrak{a}^e$ is injective. Then use a.c.c..

7.13. Transfer the Noetherian property $B \rightarrow B_{\mathfrak{p}} \rightarrow B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$.

7.14 ([Hilbert's Nullstellensatz](#)). The hint uses a method other than Rabinowitsch trick.

7.15. Hint.

7.16. Use [Exercise 7.15](#).

7.17.

7.18.

7.19.

7.20. (i) $\Rightarrow$: Rigorous proof requires using induction on $\mathcal{F}_i$, where $\mathcal{F}_i$ is formed by applying complement i times on the topology $\tau = \mathcal{F}_0$, and satisfies $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \dots \subseteq \mathcal{F}$.

(ii) $\Rightarrow$: Use [Exercise 1.19](#).

7.21. $\Rightarrow$: Use [Exercise 7.20](#).

$\Leftarrow$: Use hint. If $\overline{E \cap X_0} \neq X_0$, $E \cap \overline{E \cap X_0} \in \mathcal{F}$ by minimality, but $E \cap \overline{E \cap X_0} = E \cap X_0$. If $U_0 \subseteq E \cap X_0$, let $U_0 = U \cap X_0$ where U is open in X , then $E \cap X_0 = U_0 \cup (E \cap U^c \cap X_0)$.

7.22. If $U_0 \subseteq E \cap X_0$, use the notations similar to [Exercise 7.21](#). $U^c \cap X_0$ is closed, so by minimality $E \cap U^c \cap X_0$ is open.

7.23. Hint.

7.24. Hint.

7.25. Use [Exercises 5.11](#) and [7.24](#).

7.26.

7.27.

8 Artinian Rings

8.1. Use [Proposition 7.16](#) on local ring $A_{\mathfrak{p}_i}$. Then apply the contraction. If $\mathfrak{q}_i$ is isolated, then $\mathfrak{p}_i$ is minimal.

8.2. (i) $\Rightarrow$ (ii) Use [Proposition 8.1](#).

(iii) $\Rightarrow$ (i) Use [Proposition 8.5](#).

8.3. (i) $\Rightarrow$ (ii) By [Proposition 8.7](#), we can assume that A is an Artinian local ring. By [Proposition 8.4](#), $\mathfrak{m}^n = 0$ for some n , so we have a chain $A = \mathfrak{m}^0 \supseteq \mathfrak{m}^1 \supseteq \cdots \supseteq \mathfrak{m}^n = 0$.

By [Zariski's lemma](#), $A/\mathfrak{m}$ is a finitely generated k -module. By [Proposition 8.5](#), A is Noetherian, so $\mathfrak{m}^i$ is a finitely generated A -module, and therefore $\mathfrak{m}^i/\mathfrak{m}^{i+1}$ is a finitely generated $A/\mathfrak{m}$ -module. Use

$$\begin{aligned} \dim_k(\mathfrak{m}^i/\mathfrak{m}^{i+1}) &= \dim_{A/\mathfrak{m}}(\mathfrak{m}^i/\mathfrak{m}^{i+1}) \dim_k(A/\mathfrak{m}); \\ \dim_k(\mathfrak{m}^i) &= \dim_k(\mathfrak{m}^{i+1}) + \dim_k(\mathfrak{m}^i/\mathfrak{m}^{i+1}), \end{aligned} \quad \text{where } 0 \rightarrow \mathfrak{m}^{i+1} \rightarrow \mathfrak{m}^i \rightarrow \mathfrak{m}^i/\mathfrak{m}^{i+1} \rightarrow 0$$

to finish the proof.

(ii) $\Rightarrow$ (i) A is a k -vector space of finite dimension. Use [Proposition 6.10](#).

8.4. (i) $\Rightarrow$ (iii) The finitely-generated property is preserved for localization and quotient of modules.

(ii) $\Rightarrow$ (iii) $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$ is a finitely generated $k(\mathfrak{p})$ -algebra. By [Hilbert's basis theorem](#), any finitely generated $k(\mathfrak{p})$ -algebra is Noetherian. Then use [Exercise 8.2](#).

(iii) $\Rightarrow$ (ii),(iv) $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$ is Noetherian and integral over the field $k(\mathfrak{p})$. If $\mathfrak{q} \in B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$ is prime, $(B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}})/\mathfrak{q}$ is an integral domain, so it is a field by [Proposition 5.7](#). Hence, every prime ideal in $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$ is maximal, which means $\dim(B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}) = 0$. Thus, $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$ is Artinian by [Proposition 8.5](#). Then use [Exercise 8.2](#).

A counter example is $f : \mathbb{Q} \hookrightarrow \overline{\mathbb{Q}}$ where $\overline{\mathbb{Q}}$ is an algebraic closure of $\mathbb{Q}$.

8.5.

8.6. Consider $(A/\mathfrak{q})_{\overline{\mathfrak{p}}}$, which is an Artinian local ring. Use the following claims then [Proposition 6.7](#).

- If A is an Artinian local ring, $\mathfrak{m}$ its only maximal ideal. Then every ideal in A is $\mathfrak{m}$ -primary. Radical is the intersection of the only prime ideal.
- There is a 1-1 correspondence between $\mathfrak{p}$ -primary ideals in A and $S^{-1}\mathfrak{p}$ -primary ideals in $S^{-1}A$. Use [Proposition 3.11\(i\)](#) and [Proposition 4.9](#).

9 Discrete Valuation Rings and Dedekind Domain

10 Completion

10.1. Some completions are calculated as follows

- $A/p^n A = \bigoplus_{k \geq 0} \mathbb{Z}/p\mathbb{Z}$, where the transition maps are all identity.
- $A_n = \alpha^{-1}(p^n B) = \bigoplus_{k > n} \mathbb{Z}/p\mathbb{Z}$, so $A/A_n = \bigoplus_{k=0}^n \mathbb{Z}/p\mathbb{Z}$. The transition maps mean, if $(x^{(1)}, x^{(2)}, \dots) \in \varprojlim A/A_n$, then $x_i^{(i)} = x_i^{(i+1)} = \dots$ for all i .
- $B/p^n B = (\bigoplus_{k=0}^n \mathbb{Z}/p^k \mathbb{Z}) \oplus (\bigoplus_{k > n} \mathbb{Z}/p^n \mathbb{Z})$. The transition maps are similar to A/A_n .

The sequence $0 \rightarrow A \xrightarrow{\alpha} B \xrightarrow{p} pB \rightarrow 0$ is exact. Since p -adic topology and induced topology on pB coincide, by [Corollary 10.3](#), $\ker(\hat{p}) = \varprojlim A/A_n$.

10.2. The explicit form of the given exact sequence is

$$0 \rightarrow \bigoplus_{k > n} \mathbb{Z}/p\mathbb{Z} \rightarrow \bigoplus_{k \geq 0} \mathbb{Z}/p\mathbb{Z} \rightarrow \bigoplus_{k=0}^n \mathbb{Z}/p\mathbb{Z} \rightarrow 0.$$

Use [Exercise 10.1](#), and note that $\varprojlim \bigoplus_{k > n} \mathbb{Z}/p\mathbb{Z} = 0$. [The definition of \$\varprojlim^1 A\$ in this book is here.](#)

10.3. Denote $N = \bigcap_{\mathfrak{m} \supseteq \mathfrak{a}} \ker(M \rightarrow M_{\mathfrak{m}})$. Let $x \in \bigcap_{n=1}^{\infty} \mathfrak{a}^n M$. By [Proposition 10.17](#), we have $(1+a)x = 0$. If $\mathfrak{m} \supseteq \mathfrak{a}$, $1+a \in A - \mathfrak{m}$, so $x \in N$. Since $(\ker(M \rightarrow M_{\mathfrak{m}}))_{\mathfrak{m}} = 0$, we have $N_{\mathfrak{m}} = 0$ for all $\mathfrak{m} \supseteq \mathfrak{a}$. By [Exercise 3.14](#), $N = \mathfrak{a}N$, and therefore $N = \mathfrak{a}N = \mathfrak{a}^2 N = \dots$. Thus, $N = \bigcap_{n=1}^{\infty} \mathfrak{a}^n N \subseteq \bigcap_{n=1}^{\infty} \mathfrak{a}^n M$.

We can see N is also equal to $\bigcap_{\mathfrak{p} \supseteq \mathfrak{a}} \ker(M \rightarrow M_{\mathfrak{p}})$. Therefore, $\text{Supp}(M) \cap V(\mathfrak{a}) = \emptyset$ iff $N = M$, iff $M = \bigcap_{n=1}^{\infty} \mathfrak{a}^n M$, iff $M = \mathfrak{a}M$. By [Exercise 2.2](#), [Proposition 10.13](#) and [Proposition 10.15\(iii\)](#), $\hat{M}/\hat{\mathfrak{a}}\hat{M} = \hat{A}/\hat{\mathfrak{a}}\hat{A} \otimes_{\hat{A}} \hat{M} = A/\mathfrak{a} \otimes_A (M \otimes_A \hat{A}) = M/\mathfrak{a}M \otimes_A \hat{A}$. Since $\hat{A} \neq 0$, by [Exercise 2.3](#), $M/\mathfrak{a}M = 0$ iff $\hat{M}/\hat{\mathfrak{a}}\hat{M} = 0$. By [Proposition 10.15\(iv\)](#) and [Nakayama's lemma](#), $\hat{M} = \hat{\mathfrak{a}}\hat{M}$ iff $\hat{M} = 0$.

10.4. x not a zero-divisor, so $A \xrightarrow{x} A$ is injective. Use [Proposition 10.14](#) and apply the functor $\hat{A} \otimes_A -$.

Counter example is $k[x, y]/f$, where $f = y^2 - x^2 - x^3$. The completion is $k[[x, y]]/f$. f is irreducible in $k[x, y]$, but $f = (y - x\sqrt{1+x})(y + x\sqrt{1+x})$ is 0 in $k[[x, y]]/f$.

10.5. We can deduce from [Proposition 10.2](#) that taking $\mathfrak{a}$ -adic completion on $0 \rightarrow b^m M \rightarrow M \rightarrow M/b^m M \rightarrow 0$ preserves exactness. Thus, $(M/b^m M)^{\mathfrak{a}} \cong M^{\mathfrak{a}}/b^m M^{\mathfrak{a}}$, and therefore

$$(M^{\mathfrak{a}})^{\mathfrak{b}} = \varprojlim_m \left(\varprojlim_n (M/\mathfrak{a}^n M)/(\mathfrak{b}^m (M/\mathfrak{a}^n M)) \right) = \varprojlim_m \left(\varprojlim_n M/(\mathfrak{a}^n + \mathfrak{b}^m) M \right).$$

On the other hand, we have $(\mathfrak{a} + \mathfrak{b})^{2n} \subseteq \mathfrak{a}^n + \mathfrak{b}^n \subseteq (\mathfrak{a} + \mathfrak{b})^n$. If $x \in (\mathfrak{a} + \mathfrak{b})^n$, $x + \mathfrak{a}^n + \mathfrak{b}^n \subseteq (\mathfrak{a} + \mathfrak{b})^n$, which means each set in the basis $\{(\mathfrak{a} + \mathfrak{b})^n\}$ is open in the topology generated by the basis $\{\mathfrak{a}^n + \mathfrak{b}^n\}$, and vice versa. Thus, they generate a same topology, and therefore

$$M^{\mathfrak{a}+\mathfrak{b}} = \varprojlim M/(\mathfrak{a} + \mathfrak{b})^n M = \varprojlim M/(\mathfrak{a}^n + \mathfrak{b}^n) M.$$

We construct two homomorphisms $\varprojlim_m \left(\varprojlim_n M/(\mathfrak{a}^n + \mathfrak{b}^m) M \right) \xrightleftharpoons[\Psi]{\Phi} \varprojlim M/(\mathfrak{a}^n + \mathfrak{b}^n) M$. When $k \geq \max(m, n)$, $\mathfrak{a}^k + \mathfrak{b}^k \subseteq \mathfrak{a}^n + \mathfrak{b}^m$, so there is a canonical quotient map $\psi_{nm} : M/(\mathfrak{a}^k + \mathfrak{b}^k) M \rightarrow M/(\mathfrak{a}^n + \mathfrak{b}^m) M$. Let $\Phi((x)_{nm}) = (x)_{kk}$ be the diagonal map, and let $\Psi((y)_k) = \psi_{nm}((y)_k)$.

10.6. If $\mathfrak{a} \subseteq \mathfrak{J}$, let $x \notin \mathfrak{m}$. We can find an open neighborhood $x + \mathfrak{a}$ of x such that $(x + \mathfrak{a}) \cap \mathfrak{m} = \emptyset$.

By [Proposition 10.1](#), the closure of $\mathfrak{m}$ is $\bigcap_{n \geq 1} (\mathfrak{m} + \mathfrak{a}^n)$. If $\mathfrak{a} \not\subseteq \mathfrak{m}$, we have $\mathfrak{m} + \mathfrak{a}^n = A$, otherwise $\mathfrak{a} \subseteq \sqrt{\mathfrak{a}^n} \subseteq \mathfrak{m}$.

10.7. Let M be a finitely generated A -module. By [Proposition 10.13](#), $\hat{M} = M_{\hat{A}}$, so $\ker(M \rightarrow M_{\hat{A}}) = \bigcap_{n=1}^{\infty} \mathfrak{a}^n M$.

If $\mathfrak{a} \subseteq \mathfrak{J}$, every element in $1 + \mathfrak{a}$ is a unit, and by [Proposition 10.17](#), $\bigcap_{n=1}^{\infty} \mathfrak{a}^n M = 0$. If N is an arbitrary A -module and $x \in \ker(N \rightarrow N_{\hat{A}})$, then $x \in \ker(Ax \rightarrow (Ax)_{\hat{A}})$ and Ax is finitely generated.

If $\mathfrak{a} \not\subseteq \mathfrak{m}$, we have $\mathfrak{m} + \mathfrak{a}^n = A$. $(A/\mathfrak{m})/(\mathfrak{a}^n(A/\mathfrak{m})) = A/(\mathfrak{m} + \mathfrak{a}^n) = 0$, so the $\mathfrak{a}$ -adic completion of $A/\mathfrak{m}$ is 0. Hence, $A/\mathfrak{m} \rightarrow (A/\mathfrak{m})_{\hat{A}}$ isn't injective.

10.8. By [Exercise 1.5\(i\)](#) and [Corollary 1.6\(i\)](#), the only maximal ideal of B is $\mathfrak{m} = (z_1, \dots, z_n)$.

The $\mathfrak{m}$ -adic topologies on A , B and C are respectively equal to the induced topologies from the $\mathfrak{m}$ -adic topology on $\mathbb{C}[z_1, \dots, z_n]$. Apply [Proposition 10.2](#) or [Corollary 10.3](#) to $\mathbb{C}[z_1, \dots, z_n] \subseteq A \subseteq B \subseteq C$.

By [Proposition 10.14](#), C is a flat A -algebra. Then use [Exercise 10.7](#) and [Exercise 3.17](#).

10.9. Assume that we have constructed g_k and h_k such that $f - g_k h_k \in \mathfrak{m}^k[x]$. Denote $w_k = f - g_k h_k$. Since there exist u_k and v_k such that $1 - u_k g_k - v_k h_k \in \mathfrak{m}[x]$, we have $w_k - w_k u_k g_k - w_k v_k h_k \in \mathfrak{m}^{k+1}[x]$.

Applying polynomial division algorithm, we have $w_k u_k = q_k h_k + a_k$ with $\deg a_k < \deg h_k$. Since h_k monic, the division process shows that $q_k, a_k \in \mathfrak{m}^k[x]$. Let $b_k = w_k v_k + q_k g_k$, then $b_k \in \mathfrak{m}^k[x]$. Note that $\deg w_k < n$, we have $\deg b_k < \deg g_k$ in the polynomial ring $(A/\mathfrak{m}^{k+1})[x]$.

Let $g_{k+1} = g_k + b_k$, $h_{k+1} = h_k + a_k$, then $f - g_{k+1} h_{k+1} \in \mathfrak{m}^{k+1}[x]$. Since A complete, the coefficients of g_k and h_k converge, so we have $g, h \in A[x]$ such that $g \equiv g_k \pmod{\mathfrak{m}^k}$ and $h \equiv h_k \pmod{\mathfrak{m}^k}$. Therefore, $f - gh \in \bigcap_{k=1}^{\infty} \mathfrak{m}^k[x]$. By [Proposition 10.17](#), $\bigcap_{k=1}^{\infty} \mathfrak{m}^k[x] = 0$.

10.10. (i) $\bar{f}(x) = (x - \alpha)\bar{g}(x)$.

(ii) $\mathbb{Z}_7/7\mathbb{Z}_7 = \mathbb{Z}/7\mathbb{Z}$ by [Proposition 10.15\(iii\)](#). we also have $(x^2 - \bar{2}) = (x + \bar{3})(x - \bar{3})$ in $\mathbb{Z}/7\mathbb{Z}$.

(iii) $A = k[[x]]$ is a local ring with $\mathfrak{m} = (x)$. $\bar{f}(\bar{0}, y) = (y - \bar{a})\bar{g}(\bar{0}, y) \in A[y]$, so $a \in A$. Lift back by [Exercise 10.9](#) and let $y(x) = a$.

10.11. Let A be the ring of germs of $C^\infty(\mathbb{R})$ functions. If $f(0) \neq 0$, then $f(x)$ is always not 0 in A , so $\mathfrak{m} = \{f(x) \in A : f(0) = 0\}$ is its only maximal ideal.

$\mathfrak{m}^n = \{f(x) \in A : f^{(0)}(0) = \dots = f^{(n-1)}(0) = 0\}$, so $A/\mathfrak{m}^n \cong \mathbb{R}[x]/(x^n)$ and therefore $\hat{A} = \mathbb{R}[[x]]$. By Borel's theorem, $A \rightarrow \hat{A}$ is surjective, so $\hat{A}$ is a finitely generated A -module.

If A is Noetherian, by [Proposition 10.17](#), $\bigcap_{n=1}^{\infty} \mathfrak{m}^n = 0$, but $e^{-1/x^2} \in \bigcap_{n=1}^{\infty} \mathfrak{m}^n$.

10.12. By [Proposition 10.14](#), $A[[x_1, \dots, x_n]]$ is flat over $A[x_1, \dots, x_n]$, but from [Exercise 2.5](#) and [Exercise 2.8\(ii\)](#), $A[x_1, \dots, x_n]$ is flat over A . Then use [Exercise 2.8\(ii\)](#) again.

11 Dimension Theory

11.1. Move P to the origin to remove the constant term of f . $\mathfrak{m} = (x_1, \dots, x_n)/(f)$, so $\mathfrak{m}/\mathfrak{m}^2 = (x_1, \dots, x_n)/((x_1, \dots, x_n)^2 + (f))$. $\{x_i\}_{i=1}^n$ is a set of generators $\mathfrak{m}/\mathfrak{m}^2$. If $f \in (x_1, \dots, x_n)^2$, $\{x_i\}_{i=1}^n$ is a basis. If $f \notin (x_1, \dots, x_n)^2$, the existence of linear term of f means $\{x_i\}_{i=1}^n$ is linearly dependent.

Use [Corollary 11.18](#) on $A_{\mathfrak{m}} = k[x_1, \dots, x_n]_{\mathfrak{m}}/(f)_{\mathfrak{m}}$, then [Theorem 11.22](#).

11.2.

11.3.

11.4.

11.5.

11.6. From [Exercise 4.7\(ii\)](#), if $\mathfrak{p}_0 \subsetneq \dots \subsetneq \mathfrak{p}_n$ is a chain in A , then $\mathfrak{p}_0[x] \subsetneq \dots \subsetneq \mathfrak{p}_n[x] \subsetneq \mathfrak{p}_n[x] + (x)$ is a chain in $A[x]$.

Let $f : A \rightarrow A[x]$ be the embedding. The fiber of f^* at $\mathfrak{p}$ is $\text{Spec}(k(\mathfrak{p}) \otimes_A A[x]) = \text{Spec}(k(\mathfrak{p})[x])$. Since $\dim(k(\mathfrak{p})[x]) = 1$, a chain in $A[x]$ whose contraction is $\mathfrak{p}$ contains no more than 2 primes, so a chain in $A[x]$ contains at most $2 + 2 \dim A$ prime ideals.

11.7. Localize at $\mathfrak{p}$, we denote $m = \text{height } \mathfrak{p} = \dim A_{\mathfrak{p}}$. By [Theorem 11.14](#), there is a $\mathfrak{p}$ -primary ideal $\mathfrak{q}$ generated by m elements, so $\mathfrak{q}[x]$ also generated by m elements. From [Exercise 4.7\(iii\)](#), $\mathfrak{q}[x]$ is $\mathfrak{p}[x]$ -primary, then use [Theorem 11.14](#) again we have $\text{height } \mathfrak{p}[x] \leq m$. Then use [Exercise 11.6](#).